OpenJFEM

Reserve Factor Calculation Methods in JFEM Boxer

Internal load recovery and material and buckling assessments

Methods and implementation reference

3 November 2026

Reserve factors connect finite element recovery with defined material and instability capacities. This report explains the implemented equations, physical idealizations, numerical methods and assumptions needed to interpret those results.

The report covers isotropic and laminate strength, stringer and isolated-member columns, free-web blades, skin panels, spatially loaded webs and whole-model buckling. It also defines LIMIT and ULTIMATE load policy, coverage checks, convergence status and fixed-load sizing estimates.

17 chapters 14 original figures 101 editable equations

Implementation snapshot reviewed on 3 November 2026

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1 Purpose and implementation scope

A reserve factor compares a defined capacity with the demand recovered from a physical finite element load case. Its numerical value is meaningful only when the capacity model, load level, physical region, coordinate system and required result coverage are all defined. JFEM Boxer combines material strength checks with several elastic instability idealizations, then identifies the governing applicable check for each property region.

This report explains the implementation reviewed on 3 November 2026. It follows the complete calculation chain from applied LIMIT loads and finite element displacement recovery to member forces, shell resultants, material checks, local buckling eigenproblems and the final RF tables. It also explains whole-model SOL105 buckling, whose finite element eigenproblem is separate from the local analytical sizing checks.

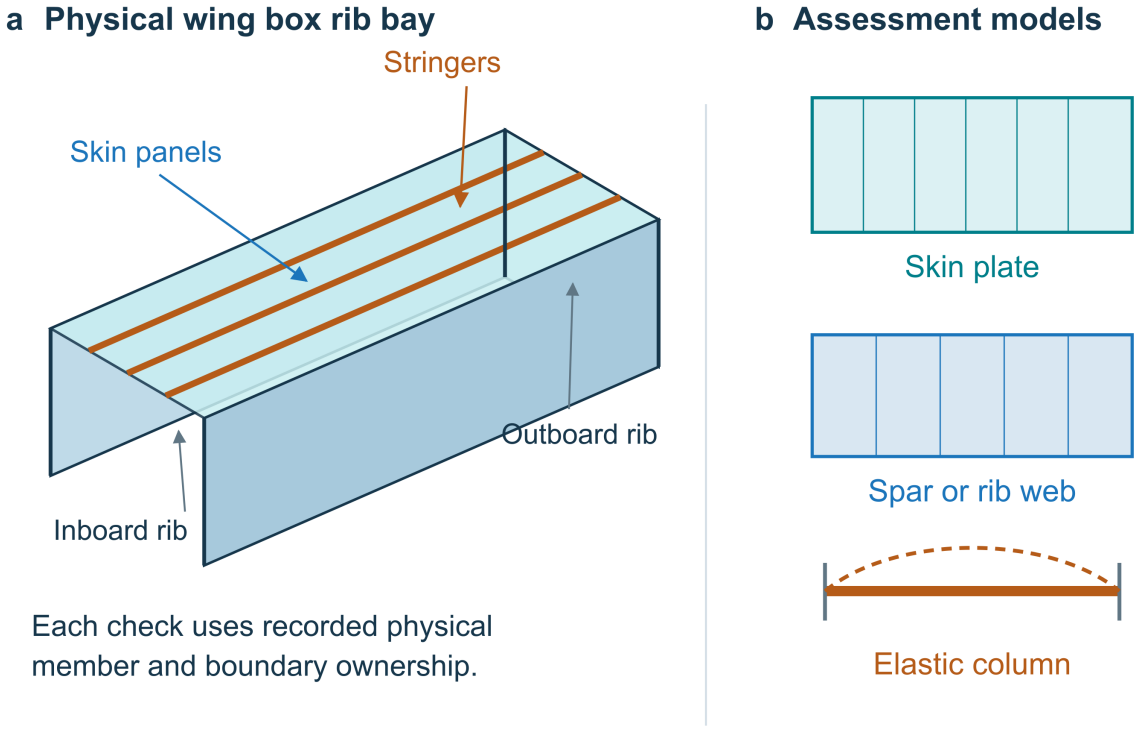


Figure 1: Physical rib bay and assessment idealizations

A generated rib bay is assessed through different physical idealizations. Skin panels, web compartments and rib-to-rib members have different load extraction and buckling laws; they are not defined solely by property IDs.

1.1 Capacities and demands refer to different physical scopes

A property ID identifies an FE material or section assignment; it does not by itself define a restrained plate or column. A stringer column may include several skin and beam properties. A skin property can contribute to more than one neighboring stringer column. A spar web or rib web compartment is bounded by physical supports and may contain several FE cells. The generated region metadata records this membership and is essential to the buckling calculations.

Material stress statistics, blade stress averages, section-cut compression and spatial web membrane fields are deliberately different demand measures. An average in one calculation must not be substituted into another. The following chapters define the measure used by each check, its sign convention and its relationship to the solved FE fields.

Table 1 Implemented local assessment families

Family	Demand and capacity	Physical scope
Isotropic material strength	Recovered equivalent stress and a material stress allowable	Complete property and selected cases
Laminate principal strain	Recovered ply-face principal strains and tensile/compressive strain limits	Eligible homogeneous orthotropic laminate property
Stringer Euler column	Compression at actual cuts and the weakest elastic section stiffness	A complete rib-to-rib stringer plus tributary skin
Stringer blade instability	Arithmetic mean signed bar stress and elastic outstand-web critical stress	PBARL T or supported blade segment between ribs
Skin panel buckling	Area-mean signed membrane resultants and equivalent rectangular plate rigidity	Skin compartment between ribs and stringers or spars
Spar web and rib web buckling	Piecewise-constant signed membrane fields and homogenized plate rigidity	Generated physical web compartment
Spar cap and rib stiffener column	Member compression and isolated-section Euler capacity	Recorded restrained member length
Rib stiffener blade instability	Mean signed axial bar stress and local free-web capacity	Recorded isotropic stiffener blade

1.2 An assessed result carries more than one number

The report data retain the governing case, failure mode, physical scope, membership, coverage and approximation status. Missing stress recovery, incomplete case coverage or an unsupported idealization is reported as unassessed. These states are not replaced by zero demand or a passing RF. Conversely, complete cases with no destabilizing demand use an explicitly identified RF 99 marker. Very large calculated RFs are displayed at a maximum of 999 while their uncapped values remain available for comparisons.

The implementation contains classical elastic plate and member models, numerical convergence checks and engineering approximations. In particular, a one hundred term plate result following failed convergence is labelled as an approximation. Its use in the combined minimum does not convert it into a converged capacity. The assumptions of every capacity model therefore belong beside its RF.

2 Load levels and reserve factor definitions

2.1 The solved physical case is the LIMIT reference

Aerodynamic forces, applied moments, engine thrust and enabled inertia forces define an operating load pattern. The new-case physical multiplier defaults to one. An explicitly entered case multiplier scales that complete pattern once; it is not the strength safety factor and does not determine the aircraft maneuver load factor. The applied FE preload already contains that case multiplier before any sizing assessment is performed.

$$\mathbf{f}_L = m_c\left(\mathbf{f}_a + \mathbf{f}_i + \mathbf{f}_t + \mathbf{f}_o\right) \tag{1}$$

Here fL is the complete LIMIT FE load vector; mc is the physical case multiplier; subscripts a, i, t and o denote aerodynamic, inertia, thrust and other applied actions.

For aircraft load matching, the aerodynamic half-wing lift target is based on the entered complete-aircraft mass and maneuver factor. The structural model remains a half lifting surface, with its own modeled structure, engine and fuel masses. The inertia loads applied to that model unload the wing through their signed accelerations. The aerodynamic target and these body forces must not be combined by counting the same wing weight twice.

$$L_{a,\mathrm{half}} = \frac{nMg}{2} \tag{2}$$

M is complete-aircraft mass in kilograms, n is the signed maneuver factor and g = 9.80665 m/s². This relation is the symmetric lift-matching convention, not a statement that the modeled half-wing mass equals M/2.

When lift matching is enabled, the application uses the corresponding signed structural and engine inertia acceleration and the actual case fuel mass. Other explicitly entered acceleration components remain part of the physical case. The case multiplier then scales the assembled loads once. Fuel fill, non-structural mass and the structural mass representation affect the physical mass and inertia model; an RF policy edit changes none of them.

2.2 ULTIMATE demand for material strength

The shared strength safety factor s defaults to 1.5. The material check compares the allowable with a factored ULTIMATE demand. For isotropic strength the demand is the chosen equivalent-stress envelope; for eligible laminate checks it is the recovered tensile or compressive principal strain. Stress and strain allowables retain their entered physical units.

$$d_U = s\, d_L, \qquad RF_s = \frac{d_{\mathrm{allow}}}{s\, d_L} \tag{3}$$

dL denotes the recovered LIMIT stress or strain demand, dU the corresponding factored strength demand and dallow its compatible material allowable.

The code preserves the recovered physical stress and strain values. Factoring belongs to the assessment, not to the stored solver arrays. Thus a property table can show a LIMIT stress of 100 MPa and, with a 300 MPa allowable and safety factor 1.5, a strength RF of 2. The material-utilization field and element inspection use the same factor.

2.3 A separate policy sets the allowed buckling load level

The buckling policy factor fb defaults to one. It defines the demand level at which a selected elastic instability capacity is assessed. The strength safety factor does not enter an Euler, blade, skin, web or other local buckling check. In a combined minimum, strength and stability RFs retain their respective denominators.

$$Q_b = f_b Q_L, \qquad RF_b = \frac{Q_{\mathrm{cr}}}{f_b Q_L} \tag{4}$$

Qcr and QL must be compatible critical and LIMIT demands: an axial force, an equivalent stress or the amplitude of a complete membrane load pattern, according to the selected model.

$$\alpha_{\mathrm{cr}} = \frac{Q_{\mathrm{cr}}}{Q_L}, \qquad RF_b = \frac{\alpha_{\mathrm{cr}}}{f_b} \tag{5}$$

At the default policy fb = 1, buckling at LIMIT gives RF = 1 and buckling at 1.5 times LIMIT gives RF = 1.5. With fb = 1.2, the latter capacity gives RF = 1.25.

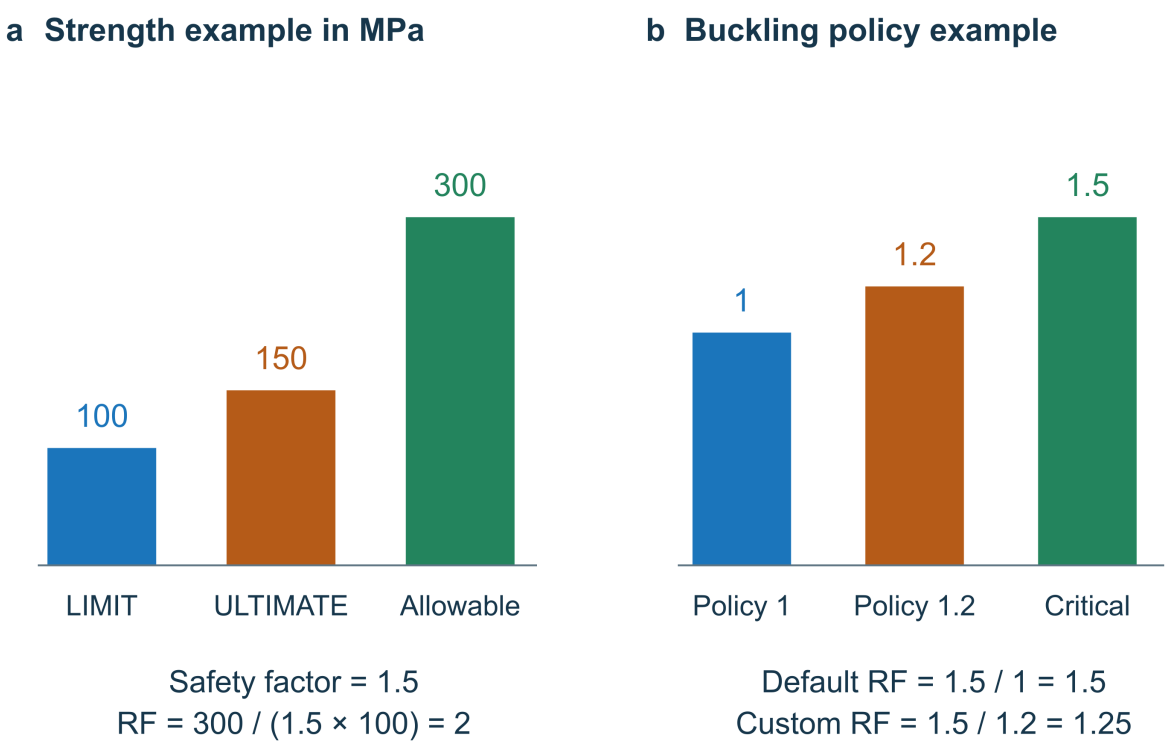


Figure 2: Independent strength and buckling load levels

Strength and buckling apply independent assessment factors to the recovered LIMIT state. The illustrated strength case has 100 MPa at LIMIT and a 300 MPa allowable, giving RF 2 with safety factor 1.5. A component buckling at 1.5 times LIMIT has RF 1.5 with the default policy 1, or 1.25 with policy 1.2. These are proportional-demand examples; SOL106 is explicitly labelled an engineering estimate rather than a new nonlinear ULTIMATE equilibrium.

2.4 Load policy and restraint are independent inputs

An engineering fixity coefficient changes an idealized capacity. For example, the Euler coefficient c multiplies the critical load, while fb multiplies its assessment demand. Plate fixity similarly scales the plate bending energy in the implemented model. The free-web blade check instead uses its fixed outstand coefficient. None of these restraint descriptions is the strength safety factor.

$$RF_{\text{Euler}} = \frac{c\pi^2 (EI)_{\min}}{f_b L^2 N_{L,c}} \tag{6}$$

NL,c is positive LIMIT compression. A change to c concerns the assumed restraint; a change to fb concerns the accepted load level. The actual FE support constraints are separate again.

Table 2 Saved sizing policy inputs

Study key	Default	Effect
assessment.strength_safety_factor	1.5	Factor on recovered strength demands
assessment.buckling_policy_factor	1	Independent factor on buckling assessment demands

Study key	Default	Effect
loads.load_factor	1 for a new case	Physical multiplier already included in the solved case

2.5 Saved loading conventions and nonlinear results

Both assessment factors must be finite and strictly positive. They are global Study inputs under Analysis and are shown in Stress Report. Their values and any legacy loading note survive Study and TOML saving. Policy-only changes refresh assessments without dirtying the FE mesh or changing the load case, native eigenvalues or raw sensitivity objectives.

Existing case multipliers are preserved. Older definitions may already have used 1.5 as an ultimate-only conversion. That conversion must be reviewed before applying the separate strength safety factor. The application flags this convention rather than silently dividing authored loads. An imported deck or result archive uses its complete analyzed preload as the LIMIT reference, so its source loading must be interpreted consistently.

For linear SOL101, proportional stress scaling is exact for the fixed solved stiffness and load direction. For SOL106, multiplying recovered full-load LIMIT demands is an engineering demand estimate. It does not recompute the nonlinear displacement, stress redistribution, follower-load direction or equilibrium at ULTIMATE load. The report labels this distinction; its RF is not a predicted nonlinear failure-load multiplier.

3 From applied loads to recovered internal actions

A reserve factor does not act directly on the aircraft lift distribution or on a list of applied nodal forces. The finite element solution first determines the structural deformation under a complete physical load case. Element routines then recover stresses, strains and internal force resultants from that state. The analytical assessments use particular parts of these recovered fields, together with the actual physical membership of a panel or member. Understanding these successive operations is essential when investigating an unexpected RF.

3.1 Applied actions internal resultants and reactions

The applied load pattern can contain aerodynamic forces and couples, prescribed forces, thrust, structural inertia and fuel inertia. An explicit case multiplier scales this complete pattern once. It belongs to the definition of the physical LIMIT case. The strength safety factor and buckling policy factor are separate assessment settings; neither is inserted into the finite element force vector by the RF routines. Therefore a change in policy must not be interpreted as a new equilibrium calculation.

Internal actions are different objects. A CBAR section force is an axial force, shear force, torque or bending moment transmitted across a member section. A shell membrane resultant is a force per unit edge length, obtained by integrating an in-plane stress through thickness. A shell bending resultant is a moment per unit edge length. A support reaction is a constrained component of the equilibrium residual. These quantities have different dimensions and different physical meanings even when their field names all contain the word force.

Table 3 The units distinguish nodal actions from shell resultants and material demands

Quantity	Meaning	Typical SI unit
Applied nodal force or reaction	External action or constrained equilibrium residual	N
Bar axial or transverse force	Internal section resultant	N
Bar torque or bending moment	Internal section couple	N m
Shell Nx Ny Nxy and Qx Qy	Internal force per unit edge length	N/m
Shell Mx My Mxy	Internal moment per unit edge length	N m/m

Quantity	Meaning	Typical SI unit
Stress and engineering strain	Local material demand	Pa and dimensionless

3.2 The physical state supplied to recovery

For a linear static case the usual free and prescribed displacement partition gives the following equilibrium relation. The subscript f denotes free independent coordinates and c prescribed coordinates. This equation explains the state from which recovery begins; it is not intended to replace the program's full constraint implementation, which also handles dependent coordinates, RBE relationships, coordinate transformations and other selected solution options.

$$\mathbf{K}_{ff}\mathbf{u}_f = \mathbf{F}_f - \mathbf{K}_{fc}\mathbf{u}_c \tag{7}$$

Linear constrained equilibrium partition; conceptual starting point, not an algorithm claim for every backend.

The solver assembles the selected load combination, resolves thermal contributions where applicable and transforms BASIC-coordinate actions into the nodal analysis coordinates. Dependent RBE3 force components are redistributed through the stored dependency coefficients. Once equilibrium is solved, the recovered displacement state is expanded into physical GRID coordinates. The recovery function _build_results_from_state constructs separate result containers for displacements, SPC forces, element forces, stresses and strains.

SPC reactions come from the appropriate constrained components of the equilibrium residual and are reported in BASIC axes. They are not the source of the stringer, skin or web demands used below. Similarly, summing the applied nodal forces on a few visible elements does not reproduce the membrane tensor or section load after the surrounding structure has redistributed the loading.

For nonlinear cases, the recorded physical state and its formulation remain important. The corotational shell route can pass element distortion displacements into shell recovery so that a rigid movement is separated from deformational strain. Other routes recover through their selected element laws using the supplied state. The RF layer does not convert these results into a plastic failure model, a universal finite-strain stress measure or an independently solved nonlinear ULTIMATE response.

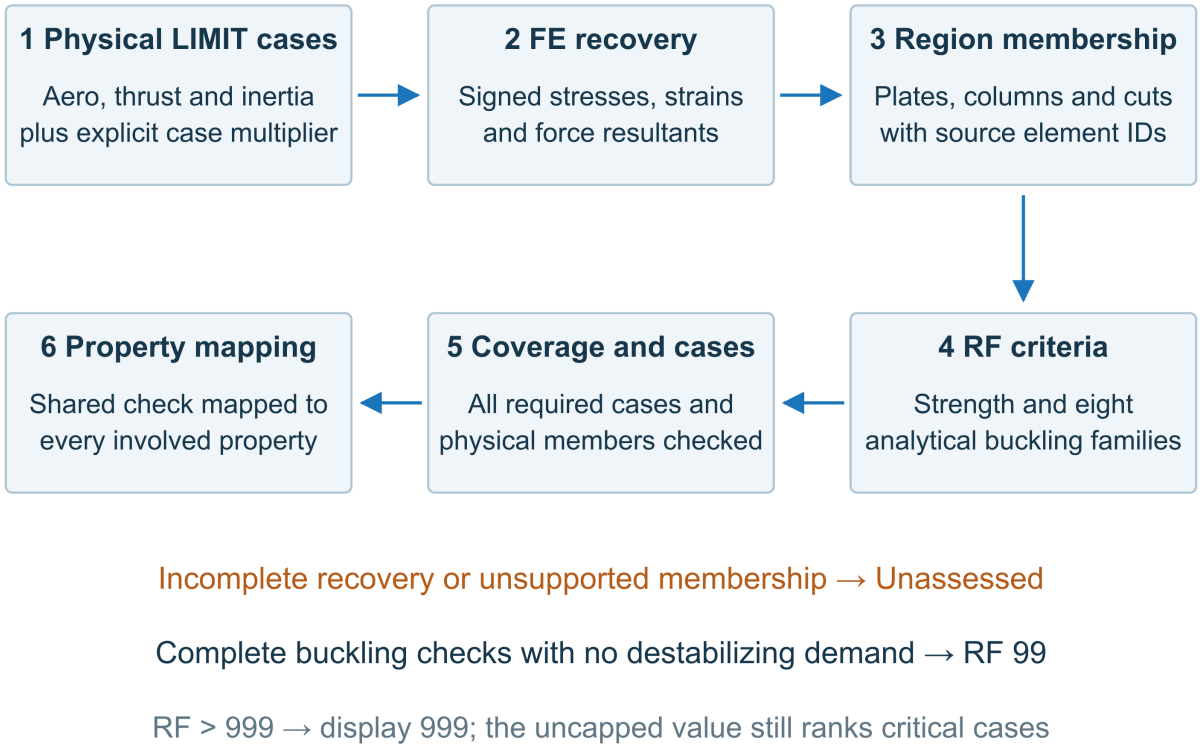


Figure 3: From FE recovery to a complete property RF

The RF calculation follows a chain from a physical case through element recovery, verified membership, criterion evaluation and coverage checks. Different criteria extract different demands from the recovered data.

3.3 The source record used by the application

The application consumes explicit element records indexed by native element ID. Each record carries its element family and recovered fields. The element_results_payload function transfers these records and supplies their coordinate and unit descriptions. It does not spread a missing element result across its neighbors or replace it by zero. The property report subsequently works with these EID records, not a visually smoothed contour.

An auxiliary native stresses[eid] scalar map is not a universal von Mises field. Depending on the element branch it can contain a principal stress, an absolute bar axial stress or a laminate VM summary. The RF calculation instead uses the explicit stress, strain and force dictionaries and the equivalent-stress recovery metadata. Reading only the convenient scalar map would therefore be an incorrect reconstruction of the implemented method.

4 Shell section recovery frames and laminate plies

4.1 Generalized strain and the shell section law

The in-plane shell strain vector contains two normal strains and engineering shear strain. Its third component is gamma_xy, which is twice the off-diagonal component of the strain tensor. This convention is used in the constitutive matrices, strain rotations and principal-strain calculation. Mixing engineering and tensor shear introduces a factor-of-two error precisely in cases where shear is important.

$$\boldsymbol{\epsilon}^0 = [\epsilon_x \quad \epsilon_y \quad \gamma_{xy}]^T, \qquad \epsilon_{xy} = \gamma_{xy}/2 \tag{8}$$

Engineering versus tensor shear convention.

For a simple flat element, the membrane strain follows derivatives of the in-plane displacements and the curvature follows derivatives of the rotational degrees of freedom. In the actual implementation, the membrane and bending operators also depend on the selected formulation, projected geometry, director or normal treatment, offsets and internal condensed modes. The schematic operator equation below is consequently a statement of the recovery structure, rather than a claim that every element uses one identical B matrix.

$$\boldsymbol{\epsilon}^0 = \mathbf{B}_m \mathbf{u}_e, \qquad \boldsymbol{\kappa} = \mathbf{B}_b \mathbf{u}_e \tag{9}$$

Generalized membrane strain and curvature; B depends on the chosen element formulation.

For a uniform isotropic plane-stress section, the material law has the following familiar form. Here E is Young's modulus, nu Poisson's ratio and t the physical shell thickness. A has units N/m and D units N m. The membrane and bending law reduces to this expression only when the property is uniform and isotropic; the actual laminate and independent-section laws must be retained otherwise.

$$\mathbf{C} = \frac{E}{1-\nu^2}\begin{bmatrix} 1 & \nu & 0 \\ \nu & 1 & 0 \\ 0 & 0 & (1-\nu)/2 \end{bmatrix}, \quad \mathbf{A} = t\mathbf{C}, \quad \mathbf{D} = \frac{t^3}{12}\mathbf{C} \tag{10}$$

Uniform isotropic plane-stress section law.

With laminate membrane and bending coupling included, the generalized resultants are related to membrane strain and curvature through A, B and D. The native output moment convention has the opposite sign to the positive constitutive moment used in many laminate textbooks. The following equation records that convention explicitly. A sign change in a reported bending moment must not be mistaken for compressive versus tensile membrane demand.

$$\mathbf{N} = \mathbf{A}\boldsymbol{\epsilon}^0 + \mathbf{B}\boldsymbol{\kappa}, \qquad \mathbf{M}_{out} = -\left(\mathbf{B}^T\boldsymbol{\epsilon}^0 + \mathbf{D}\boldsymbol{\kappa}\right) \tag{11}$$

Shell section resultants and the native printed-moment sign convention.

4.2 Recovery locations and internal modes

It is inaccurate to describe all shell stresses as centroid stresses. In the ordinary incompatible quadrilateral branch, internal Wilson-mode amplitudes are reconstructed by static condensation. Strains and curvatures are then evaluated at Gauss points and combined using the element area measure. Compatible-center, projected and coupled-condensed routes retain their own work-conjugate center or mean contracts. Some transverse shear resultants are explicitly recovered as area means rather than extrapolated centroid values.

$$\boldsymbol{\alpha} = -\mathbf{K}_{bb}^{-1}\mathbf{K}_{ab}^T\mathbf{u}_e, \qquad \boldsymbol{\epsilon}_g = \mathbf{B}_{m,g}\mathbf{u}_e + \mathbf{B}_{i,g}\boldsymbol{\alpha} \tag{12}$$

Internal incompatible amplitudes and Gauss-point strain in the ordinary quadrilateral recovery branch.

$$\langle \boldsymbol{\epsilon} \rangle_A = \frac{\sum_g w_g |J_g| \boldsymbol{\epsilon}_g}{\sum_g w_g |J_g|} \tag{13}$$

Area-weighted generalized field; not a nodal stress smoothing operation.

In these equations alpha contains the element's internal incompatible amplitudes, not additional global nodal unknowns. The integration measure is based on the element Jacobian. Reconstructing the internal amplitudes matters because the compatible displacement field alone does not necessarily reproduce the strain field associated with the assembled stiffness. The triangle route has its own constant membrane and curvature operators and its selected transverse-shear recovery.

Ordinary isotropic stresses are recovered through the thickness from membrane strain plus curvature. Default PSHELL stress fibers lie at minus and plus half the thickness. If Z1 or Z2 has been explicitly authored, the code evaluates the same linear stress field at that distance instead. Such an output-location choice does not change the membrane or bending resultant. On a thickness edit, an implicit half-thickness fiber moves with the shell; an explicitly supplied distance remains explicit.

$$\boldsymbol{\epsilon}(z) = \boldsymbol{\epsilon}^0 + z\boldsymbol{\kappa}, \qquad \boldsymbol{\sigma}(z) = \mathbf{C}\boldsymbol{\epsilon}(z), \qquad z_{1,2} = \mp t/2 \tag{14}$$

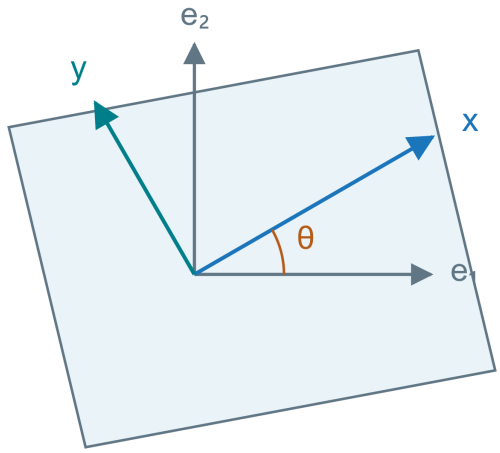
Ordinary isotropic default fiber stress law.

4.3 Reference axes and tensor transformation

Native shell recovery begins in a geometric frame determined by connectivity. A triangular local x direction follows edge 1 to 2; its normal follows the element orientation. The default quadrilateral frame uses diagonal geometry and an in-plane projection. That geometric frame does not generally coincide with the structural direction in which a panel or stringer is assessed.

For generated skins, Boxer defines result x along the local stringer segment toward the tip. Rib result x is horizontal in the rib plane and directed aft; for a line-of-flight rib it is global positive x. Spar-web x follows the nominal stringer direction projected into the web. Imported shell results retain their native geometric axes rather than being reoriented as if the imported model were a generated wing.

a Element and panel axes



$x = c\,e_1 + s\,e_2$

$y = -s\,e_1 + c\,e_2$

b Tensor transformation

$c = \cos\theta,\ s = \sin\theta$

$N_x' = c^2 N_x + s^2 N_y + 2cs\,N_{xy}$

$N_y' = s^2 N_x + c^2 N_y - 2cs\,N_{xy}$

$N_{xy}' = cs(N_y - N_x) + (c^2 - s^2) N_{xy}$

Use γ / 2 for a strain tensor; engineering shear is twice ε_{xy}.

Do not rotate individual scalar maxima: rotate the signed tensor first.

Figure 4: Element resultants transformed to panel axes

Complete signed tensors are rotated into the assessment frame before averaging or projection. A collection of scalar maxima cannot be rotated as though it were a tensor.

$$\begin{aligned} S_{x'} &= c^2 S_x + s^2 S_y + 2cs S_{xy} \\ S_{y'} &= s^2 S_x + c^2 S_y - 2cs S_{xy} \\ S_{x'y'} &= cs(S_y - S_x) + (c^2 - s^2) S_{xy} \end{aligned} \tag{15}$$

Any symmetric in-plane stress/resultant tensor rotation.

The tensor S in this transformation can represent in-plane stress, membrane force or printed bending moment. The axis convention is x prime equal to c times the original first axis plus s times the original second axis. Transverse shear resultants rotate as a two-component vector. If any component of a required tensor is missing, the application does not relabel the remaining components as a complete rotated field.

$$\begin{aligned} \epsilon_1 &= c^2 \epsilon_x + s^2 \epsilon_y + cs\gamma_{xy} \\ \epsilon_2 &= s^2 \epsilon_x + c^2 \epsilon_y - cs\gamma_{xy} \\ \gamma_{12} &= 2cs(\epsilon_y - \epsilon_x) + (c^2 - s^2)\gamma_{xy} \end{aligned} \tag{16}$$

Engineering strain rotation.

This second transformation is written specifically for engineering strain. It is equivalent to dividing gamma by two, rotating the symmetric strain tensor and doubling the resulting tensor shear. Explicit physical ply-face records have already been expressed in each ply's own material axes and are deliberately not rotated again by the subsequent geometric-to-material output transformation.

4.4 Classical laminate integration and explicit ply faces

For a MAT8 ply, the reduced plane-stress stiffness is built from E1, E2, nu12 and G12. Reciprocal Poisson coupling determines nu21. The rotated stiffness Qbar is integrated through the actual original ply boundaries to form A, B and D. These expressions are the constitutive theory implemented in ModelBuilder and are consistent with the classical laminate framework described in NASA RP 1351 [1]. They describe elasticity; that reference does not establish a composite failure criterion.

$$\nu_{21} = \nu_{12}E_2/E_1, \quad \Delta = 1 - \nu_{12}\nu_{21}, \quad \mathbf{Q} = \begin{bmatrix} E_1/\Delta & \nu_{12}E_2/\Delta & 0 \\ \nu_{12}E_2/\Delta & E_2/\Delta & 0 \\ 0 & 0 & G_{12} \end{bmatrix} \tag{17}$$

Reduced MAT8 plane-stress law before rotation into laminate axes.

$$\begin{aligned} A_{ij} &= \sum_k \overline{Q}_{ij}^{(k)}(z_k - z_{k-1}) \\ B_{ij} &= \frac{1}{2}\sum_k \overline{Q}_{ij}^{(k)}(z_k^2 - z_{k-1}^2) \\ D_{ij} &= \frac{1}{3}\sum_k \overline{Q}_{ij}^{(k)}(z_k^3 - z_{k-1}^3) \end{aligned} \tag{18}$$

Actual ply-thickness integration used for membrane, coupling and bending laminate stiffness.

Each original ply retains its material ID, angle, bottom and top z locations and constitutive information. For face recovery, the generalized membrane strain and curvature are first expressed in the common laminate frame. The in-plane strain at each physical ply face is then formed, rotated to the ply material 1 and 2 axes, and paired with the corresponding stress. Both sides of every ply are retained because a change of orientation or stiffness can make the response at an interface different in the adjacent materials.

$$\boldsymbol{\sigma}_k^L(z) = \overline{\mathbf{Q}}_k\left(\boldsymbol{\epsilon}_L^0 + z\boldsymbol{\kappa}_L\right), \quad z \in \{z_{k-1}, z_k\} \tag{19}$$

Explicit bottom and top of each physical laminate ply; stress is then rotated to ply axes.

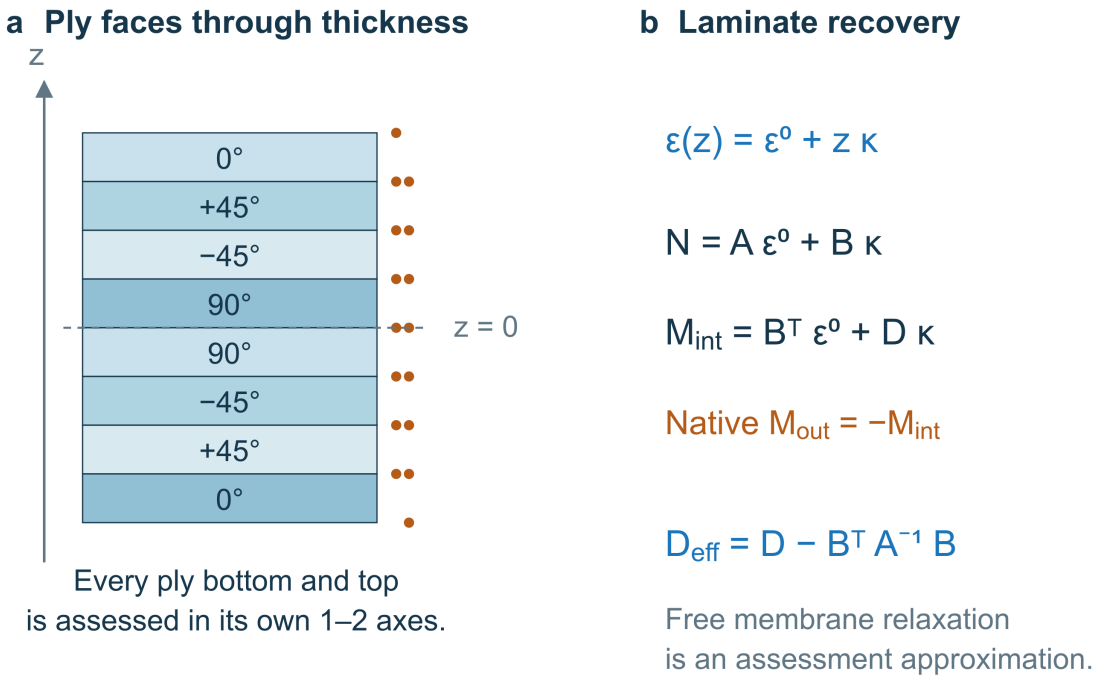


Figure 5: Laminate ply faces and section stiffness

Physical every-ply bottom and top recovery is distinct from legacy generalized strain records. The printed moment sign and the relaxed local bending law must be stated explicitly when interpreting laminate output.

The face recovery requires complete finite generalized strains, valid original ply geometry, supported material data and a coherent membrane/bending frame. Experimental settings that use separate constitutive frames do not define one physical ply frame for this purpose. The code therefore marks that face recovery unavailable instead of constructing a surrogate strain state.

Table 4 The legacy z labels must not be read as a physical surface-strain pair

Stored strain field	Actual meaning	Use in a laminate allowable check
Legacy z1 with fiber tag 0	Membrane strain vector	Not a physical top or bottom face
Legacy z2 with fiber tag minus 1	Negative curvature vector in inverse length	Not a physical top or bottom face
ply_faces bottom and top	Physical strain at both faces of each original ply in material axes	Required for the implemented principal-strain assessment

The historical PCOMP stress z1 and z2 summaries are likewise not a substitute for a complete every-ply face assessment. Those summary rows can represent first and last ply midpoint surrogates. The explicit ply_faces arrays are the fields used when the material criterion requires actual bottom and top strains for every ply.

5 Beam section and solid stress recovery

5.1 From nodal displacement to section force

Beam local x follows the effective offset end points. The orientation vector defines the remaining section axes. GRID analysis-coordinate translations and rotations are transformed into this element frame. End offsets introduce the corresponding rotation cross offset displacement, and the thermal extension is removed before the elastic section-force calculation. These operations are necessary for an eccentric stringer: using only the displacement difference between the two original GRID positions would omit part of its section deformation.

For an ordinary constant-section beam, the internal local nodal force vector is the local stiffness times the local displacement vector minus the fixed-end contribution of the active distributed loading. Releases and the selected reference or neutral-axis convention are included in that path. Station-varying PBEAM or PBEAML definitions have separate recovery logic and do not automatically qualify for a constant-section equivalent-stress envelope.

$$\mathbf{f}_e^{local} = \mathbf{K}_e^{local}\mathbf{u}_e^{local} - \mathbf{f}_{e,fixed}^{local} \tag{20}$$

Ordinary beam internal nodal force vector after active distributed fixed-end loads, with releases/offset mapping applied.

The local nodal force vector is mapped into the signed section resultants exported as axial, shear_1, shear_2, torque and the two end bending-moment pairs. Axial force is tension positive. Native axial stress is N divided by the actual section area; a varying section can retain distinct end axial stresses. Those signed axial values are used by compression-specific blade and column checks, whereas an equivalent-stress check also needs bending, shear and torsion.

5.2 Total normal stress at a physical section

The end_a and end_b normal-stress entries contain bending contributions, with axial stress stored separately. Boxer completes the bending samples at the actual T or rectangular polygon vertices, because dimension-defined T sections do not otherwise have useful native C D E F recovery points. To obtain total normal stress one must add axial stress with the correct sign. A zero bending placeholder is not evidence that the member carries no stress.

$$\sigma_{N,b}(y,z) = \frac{N}{A} - \frac{M_1 y}{I_1} - \frac{M_2 z}{I_2} \tag{21}$$

Symmetric T/BAR polygon-vertex normal stress convention used by the supported equivalent envelope.

Here the coordinates y and z, the moments M1 and M2 and the inertias I1 and I2 follow the program's section convention. The expression assumes the supported symmetric principal section axes. A user should not exchange the two inertias merely because a different textbook gives them the names Iy and Iz. The general native surface recovery also carries a product-of-inertia expression, while the supported T and rectangular equivalent envelopes explicitly require the appropriate symmetric section.

$$\sigma_{\mathrm{bound}} = \max_{e\in\{A,B\}}\max_{(y,z)\in\mathcal{P}}\left|\frac{N}{A} - \frac{M_{e1}y}{I_1} - \frac{M_{e2}z}{I_2}\right| \tag{22}$$

Both-end polygon envelope of total normal stress; maxima are not necessarily co-located with shear.

5.3 The thin walled T section equivalent envelope

The T envelope uses full flange width b, total height h, flange thickness tf and web thickness tw, with clear web height hw equal to h minus tf. Its area, centroid and inertias come from nonoverlapping finite-thickness flange and web rectangles. Eligibility requires positive geometry, a flange wider than the web, and flange thickness divided by full width and web thickness divided by clear height no greater than 0.10. These checks define the range of the thin-wall idealization.

The normal bound is the largest absolute total axial-plus-bending stress over both beam ends and all section polygon vertices. The transverse shear calculation starts at free edges and uses the VQ over I shear-flow principle, with the wall thickness converting flow to stress [5]. The code combines bounds from both transverse shear components. It also retains a torque bound for the centroid to shear-center eccentricity rather than allowing a favorable sign cancellation at an unexamined point.

$$q(s) = \frac{VQ(s)}{I}, \qquad \tau(s) = q(s)/t(s) \tag{23}$$

Thin-wall transverse shear-flow principle; exact code bounds combine both section shear components.

$$\begin{aligned} J_{thin} &= (h_w t_w^3 + b t_f^3)/3 \\ T_b &= |T| + |V_2||y_f - \overline{y}| \\ \tau_{T,w} &= T_b t_w / J, \quad \tau_{T,f} = T_b t_f / J \end{aligned} \tag{24}$$

Thin-T torsion reference and the code shear-center torque bound; an explicitly tagged 2D section can supply its own J.

For the usual thin-wall section, the torsion constant follows the sum of each wall length times its thickness cubed divided by three. If the section property is explicitly tagged with the finite-thickness two-dimensional Saint-Venant torsion formulation, its positive finite J is used instead. The surrounding stress-envelope formula still carries its stated thin-wall assumptions. Providing a more detailed torsion constant alone does not turn the output into a full local solid stress solution.

$$\sigma_{\mathrm{VM},envelope} = \sqrt{\sigma_{\mathrm{bound}}^2 + 3\tau_{\mathrm{bound}}^2} \tag{25}$$

Conservative combined normal/shear section envelope, not pointwise recovered VM.

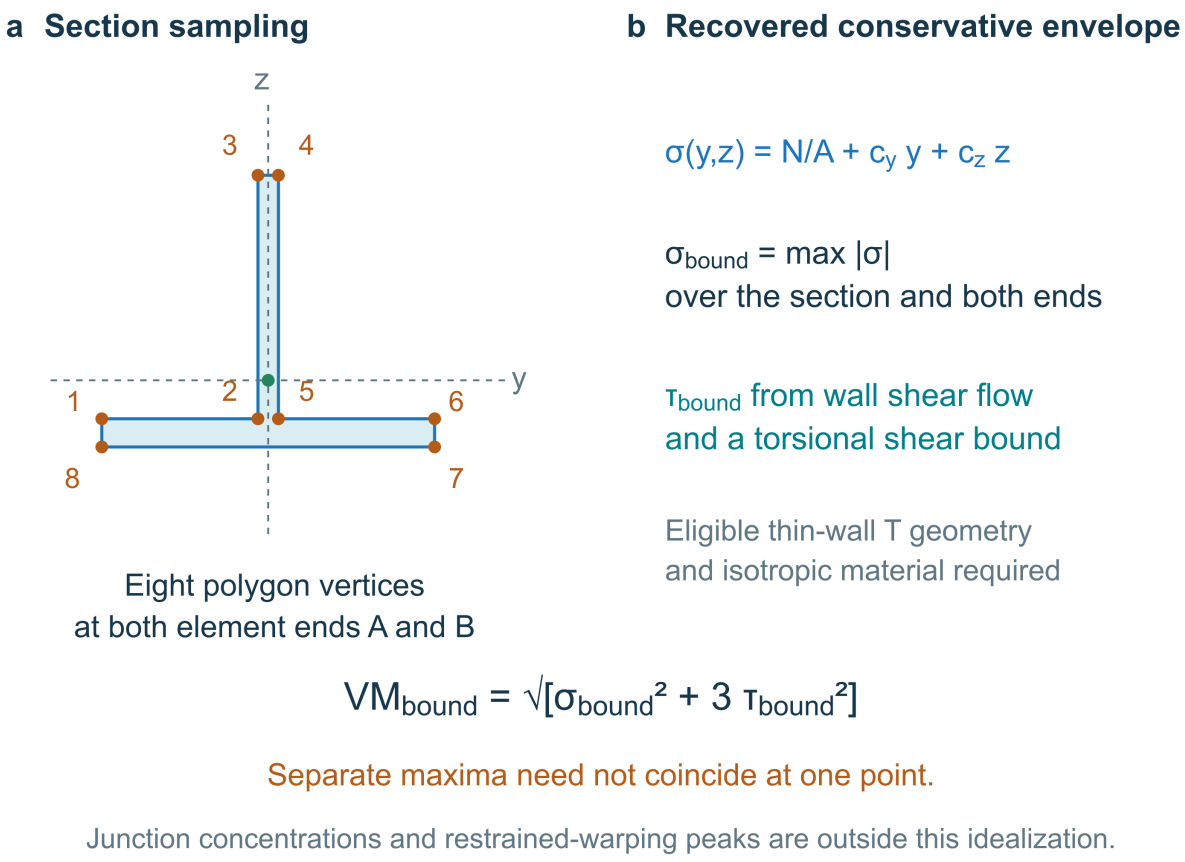


Figure 6: The conservative beam equivalent stress envelope

The supported T-section strength calculation combines a both-end polygon normal-stress bound with bounded transverse and torsional shear. Those maxima need not occur at one point, so the result is a conservative section envelope.

This last distinction is central. The code combines the largest normal bound with the largest shear bound even when their locations differ. Its VM envelope is deliberately conservative within the supported beam-section idealization; it is not a pointwise reconstruction of the section stress field. Junction concentrations, restrained warping, transverse normal stress and local plasticity are not included. The engineering user can use a shell or solid representation when those mechanisms matter.

5.4 Rectangular caps and section applicability

A solid rectangular BAR section uses the same both-end corner normal bound, a transverse shear bound proportional to the sum of the absolute shear components, and a conservative rectangular Saint-Venant torsion expression. In the equation below a is the longer side and b the shorter side. The torsion coefficient is 0.20 by code choice; a circular-shaft torque times radius divided by polar inertia is not used for the rectangle [7].

$$\tau_{V,rect} = \frac{1.5(|V_1| + |V_2|)}{A}, \qquad \tau_{T,rect} = \frac{5|T|}{ab^2}, \quad a \geq b \tag{26}$$

Solid rectangular BAR transverse and conservative torsion bounds; coefficient c=0.20.

The equivalent-stress recovery requires complete finite resultants and section dimensions consistent with the stored A, I1, I2 and J. It rejects unsupported material and shape combinations, incompatible section properties and incomplete end resultants. It never substitutes absolute axial stress for missing VM. This is why a complete Euler or blade assessment can coexist with an unavailable material equivalent-stress RF: their required recovery fields are different.

A distributed load can create an interior stress maximum even when its condensed end vector is zero. The implementation therefore checks selected load membership, including body loading, instead of relying only on the fixed-end-vector norm. Tapered or station-varying beams and active distributed beam loads need appropriate interior recovery before the constant-section envelope can be reported as complete. In a Boxer generated model, many physical loads are mapped to nodal attachments; the recorded provenance still controls applicability.

5.5 Imported volume elements

The native solid route covers supported four- and ten-node tetrahedra, eight-node hexahedra and six-node prisms. Translational nodal DOFs are transformed to global coordinates and thermal expansion is removed before the isotropic three-dimensional B u strain and material stress calculation. Centroid fields are recovered. Ordinary native CHEXA8 can also retain corner stresses, using the corresponding enhanced or assumed strain field when that formulation is active. The imported Abaqus C3D8 branch has its own recovery contract.

$$\sigma_{\mathrm{VM},3D} = \sqrt{\frac{(\sigma_x - \sigma_y)^2 + (\sigma_y - \sigma_z)^2 + (\sigma_z - \sigma_x)^2}{2} + 3(\tau_{xy}^2 + \tau_{yz}^2 + \tau_{zx}^2)} \tag{27}$$

Three-dimensional isotropic equivalent stress for supported native volume elements.

The imported solid contour is labelled centroidal and is constant on each displayed exterior face of the original volume element. The native stress row may additionally contain corner samples. The generic material VM sampler examines explicitly recovered values recursively, so its property RF can be governed by a stored corner even when the visible centroid contour is lower. No cross-element smoothing is used for this decision, and a solid PID does not by itself define one of the generated shell-panel buckling regions.

6 Material criteria and property stress envelopes

6.1 Isotropic stress and orthotropic laminate strain

The isotropic material criterion compares recovered von Mises stress with one positive allowable. For a shell this is the plane-stress expression below; solid recovery uses the three-dimensional form given previously. A PSHELL needs both recovered face values. For beams, the sampler requires an explicitly supported complete equivalent-stress recovery. A positive material allowable is a user-supplied design criterion and does not change the stiffness, density, elastic law or applied loading.

$$\sigma_{\mathrm{VM}} = \sqrt{\sigma_x^2 - \sigma_x\sigma_y + \sigma_y^2 + 3\tau_{xy}^2} \tag{28}$$

In-plane plane-stress von Mises measure.

For orthotropic laminates, the implemented material criterion is based on in-plane principal strain, not on material-direction stress or a directional interaction failure theory. The laminate must use the same MAT8 material ID in every ply, although the ply orientations can differ. Two dimensionless positive limits are entered: a tensile principal-strain limit and the positive magnitude of the compressive limit. Heterogeneous orthotropic laminates do not receive an invented common limit.

$$\epsilon_{\max,\min} = \frac{\epsilon_1 + \epsilon_2}{2} \pm \sqrt{\left(\frac{\epsilon_1 - \epsilon_2}{2}\right)^2 + \left(\frac{\gamma_{12}}{2}\right)^2} \tag{29}$$

Principal strains of the in-plane engineering-strain tensor.

The half engineering-shear term in this equation is essential. Principal strains are the eigenvalues of a symmetric two-dimensional tensor; gamma12 is not itself the tensor off-diagonal. For every element, ply and physical bottom or top face, the positive maximum principal strain is compared with the tensile limit and the magnitude of the negative minimum principal strain with the compressive limit. The greatest utilization governs. It is not averaged over plies, and a VM percentile setting does not trim these strain samples.

$$U_T = \frac{f_s \max(0, \epsilon_{\max})}{\epsilon_{T,allow}}, \quad U_C = \frac{f_s \max(0, -\epsilon_{\min})}{\epsilon_{C,allow}}, \quad RF = \frac{1}{\max U_{T,C}} \tag{30}$$

Homogeneous orthotropic laminate strength utilization; maximize over all actual element/ply/face/case samples.

Table 5 A principal strain example showing why engineering shear must be halved

Illustrative input or step	Value	Interpretation
Face engineering strains	epsilon1 0.002 epsilon2 minus 0.001 gamma12 0.004	Tensor shear is 0.002
Principal strains	0.003 and minus 0.002	Mean 0.0005 plus or minus radius 0.0025
Allowables and strength factor	Tension 0.004 compression 0.003 factor 1.5	Two material branches assessed separately
Utilizations	Tension 1.125 compression 1.000	Tension governs
RF and margin	0.88889 and minus 0.11111	Illustrative arithmetic only not an FE validation result

These are elastic pointwise demand comparisons. The implemented principal-strain rule is not Tsai-Wu, Hashin, Puck, delamination, interlaminar strength or progressive damage. In particular, a complete homogeneous laminate strain RF does not certify the absence of those other failure mechanisms.

6.2 The direct material viewer and Stress Report have different scopes

The direct Material allowable utilization view can evaluate supported MAT1 ply-face stress comparisons in a PCOMP using the material limit for each ply. The property-level Stress Report VM path, however, deliberately excludes PCOMP, and its principal-strain path only covers homogeneous same-MAT8 laminates. Thus an isotropic-ply laminate can show a direct ply utilization without having a material-strength contributor in the property Minimum RF table. This is a current coverage distinction, not an infinite strength margin.

A generated SI aluminium 2024 entry is assigned an editable 300 MPa stress default. That number is an application design default, not a certified allowable for every product form, condition or environment. Imported source-unit models do not silently receive the generated-SI default. The comparisons themselves are unit-consistent ratios, provided the recovered demand and entered allowable use the same physical unit system.

6.3 One element sample and one case statistic

For the VM property report, the first aggregation is within an element: the largest explicitly recovered finite nonnegative equivalent stress over its available faces or sample records is retained. This produces one peak per EID. The application then groups those peaks by property and by eligible physical load case. The raw maximum and its element and face are retained even if the user chooses an average or trimmed statistic for the RF.

$$s_{e,c} = \max_{recovered\ samples}\sigma_{VM,e,c}, \quad S_{p,c}^{\max} = \max_{e\in p} s_{e,c}, \quad S_{p,c}^{mean} = \frac{1}{n_p}\sum_{e\in p} s_{e,c} \tag{31}$$

One peak per element, then a property statistic separately for each case.

The arithmetic mean gives each element one equal vote. It is not weighted by surface area, volume, shell gauge, stiffness or mass. Consequently, a locally refined region contributes more samples to this statistic. That behavior is a defined reporting choice; it should not be confused with the area weighting or cut integration used to recover a physical panel or column load. Selecting a mean does not remove the physical hotspot from the model.

6.4 The percentile option means an average below the cutoff

For percentile p, the code sorts the nonnegative element peaks and interpolates linearly at zero-based rank r equal to n minus one times p over one hundred. It retains values at or below that cutoff, including roundoff-scale ties, and calculates the mean of those retained peaks. It does not report the percentile value itself, and it does not average only the highest p percent of stresses.

$$r = (n-1)p/100, \quad j = \lfloor r \rfloor, \quad f = r - j, \quad q_p = (1-f)s_{(j)} + f s_{(\lceil r \rceil)} \tag{32}$$

Linear percentile interpolation, with zero-based sorted indexing.

$$\mathcal{I}_p = \{e : s_e \le q_p\}, \qquad S_{p,c}^{trim} = \frac{1}{|\mathcal{I}_p|}\sum_{e\in\mathcal{I}_p} s_{e,c} \tag{33}$$

Mean below the chosen percentile; code also retains roundoff-scale ties.

Consider element peaks of 80, 100, 120 and 500 MPa in one property and one case. At the 75th percentile, r is 2.25 and the interpolated cutoff is 215 MPa. The retained values are 80, 100 and 120 MPa, so the mean below the cutoff is 100 MPa. The raw maximum remains 500 MPa. With a 300 MPa allowable and strength factor 1.5, maximum, full mean and trimmed mean respectively give RFs of 0.4, 1.0 and 2.0. This example demonstrates the choice of statistic; it is not evidence that the discarded peak is physically harmless.

6.5 The load case envelope follows the statistic

The selected property statistic is calculated separately in each included case, then the greatest case value is used for strength. Stresses from several cases are never pooled into one mean. In the preceding example, if another case has a selected statistic of 140 MPa, the trimmed envelope becomes 140 MPa and the RF is 300 divided by 1.5 times 140, or 1.42857. The governing label belongs to that second case even though the largest individual raw stress may belong to the first.

$$S_{p,env} = \max_c S_{p,c}, \qquad RF_{p,strength} = \frac{\sigma_{allow}}{f_s S_{p,env}}, \qquad MS = RF - 1 \tag{34}$$

Worst case of per-case statistics, not an average across cases.

A VM property needs one compatible MAT1 material and one positive common allowable. Different membrane, bending or shear materials do not receive a guessed common stress limit. The finite RF is assigned only if the selected cases and every required element are complete. For the principal-strain report, the analogous envelope is the greatest actual face utilization rather than the VM statistical envelope. The reported margin is RF minus one.

7 Physical column idealizations and restraints

A column RF belongs to a defined physical member and its restraint model. It is not obtained from the color of a stress contour. Before calculating an Euler capacity, Boxer establishes a complete unbraced path, identifies the actual adjacent structure, and constructs cross-sections from current geometry and properties. The stringer check includes tributary skin; the spar-cap and rib-stiffener checks deliberately do not. This distinction controls both stiffness and demand.

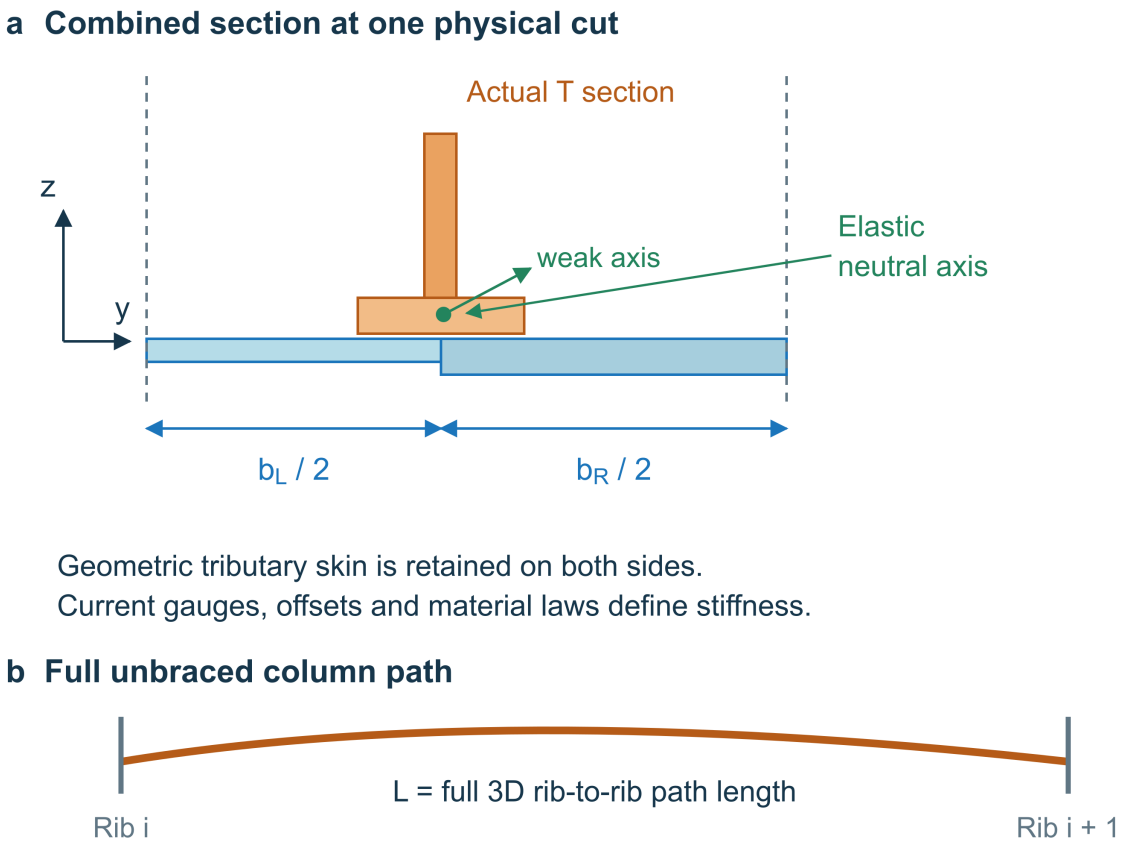


Figure 7: Tributary stringer section and rib to rib length

The stringer Euler section combines the actual stringer with half the geometric skin width to the neighboring stringer or spar on each side. This is a tributary section, not a postbuckling effective-width rule. Current material stiffness, thickness and offsets determine the elastic neutral axis and weakest bending stiffness; the unbraced length is the complete actual rib-to-rib path. The axes and green neutral-axis point are illustrative sketch coordinates, not a numerical section result or the native section-orientation convention.

7.1 Physical stringer column and tributary skin membership

For every generated stringer segment, Julia finds its actual beam path or shell-stringer attachment-edge path. The full unbraced length L is the sum of modeled three-dimensional segment lengths between the two physical ribs. It is not the nominal rib pitch, projected eta gap or length of a single bar element.

A cross-section is constructed at each current stringer path segment midpoint, in the plane normal to that local segment. Its axis is oriented tipward. The section normal follows the local outward skin normal projected perpendicular to the stringer; the remaining transverse direction completes an orthogonal section frame.

The neighboring active stringer paths, or the actual front/rear spar paths at an edge, are intersected by this plane. The included skin extends halfway from the current stringer to each neighboring support. The two half widths need not be equal.

Skin and shell-stringer faces are cut by the plane and clipped to this full half-width interval. Actual face materials, thicknesses, laminate stacks and offsets are used. A normal cut near a swept rib can enter the neighboring rib bay; the source deliberately searches those actual neighboring faces instead of extrapolating the current property.

A shell-stringer combined flange is included as its actual combined shell property, with the web belonging to the current stringer. There is no separate fictitious duplicate skin stiffness. Coverage of the entire tributary interval is checked; a gap, ambiguous polygon intersection, missing web or unresolved membership produces an unassessed criterion.

The same full L is used with each local section stiffness. The minimum capacity-to-local-compression ratio across cuts and eligible cases governs. This is an approximation for a tapered or curved member; it is not a variable-EI differential eigenproblem.

A runout whose end is not supported by the two recorded adjacent ribs can retain real section membership but has no assigned rib-to-rib Euler RF until its end restraint is defined. Geometric membership and end-restraint validity are separate facts.

$$L = \sum_{e \in \mathcal{P}} \|\mathbf{x}_{eB} - \mathbf{x}_{eA}\| \tag{35}$$

Actual modeled three-dimensional path length over the complete unbraced member.

$$y_{\text{left}} = \frac{\mathbf{e}_y \cdot (\mathbf{x}_{\text{left}} - \mathbf{x}_0)}{2}, \qquad y_{\text{right}} = \frac{\mathbf{e}_y \cdot (\mathbf{x}_{\text{right}} - \mathbf{x}_0)}{2} \tag{36}$$

Halfway tributary bounds obtained from actual adjacent support intersections in each local section.

Consider a stringer located close to a spar. Its tributary skin width is not necessarily the nominal stringer pitch divided by two on both sides. The code intersects the actual neighboring paths at the current section. If the left support is a spar and the right support is another stringer, those different distances set the two half widths. Spar perturbations, local stringer angles and sweep can all change this construction. The descriptor records the resulting widths and frame, allowing the RF to be traced to actual shell elements rather than a nominal panel area.

The section plane can cut shells from an adjacent physical rib bay when a rib is swept relative to the stringer. That is a geometric consequence, not a duplicate assignment. The algorithm searches a spatial index of real candidate faces and clips each intersection to the tributary interval. Actual property IDs, laminate laws and offsets follow those faces. It does not extend the current bay thickness into its neighbor. This avoids an erroneous capacity when a property discontinuity lies close to an oblique section.

The distinction between a physical path and the mesh subdivisions along it is essential. Increasing the number of bar elements between ribs does not reduce the Euler length. It adds more local section stations along the same complete member. Each station must still resist an instability with the full recorded support-to-support length. Consequently a mesh refinement cannot artificially raise Euler capacity through a smaller L for every new bar element.

7.2 Restraints and member eligibility

The Euler restraint coefficient is an engineering representation of end fixity. With c equal to one, the resistance corresponds to the familiar pinned-end Euler idealization. A different c is equivalent to an effective length factor K equal to one divided by the square root of c. It does not modify the actual FE support cards, nor does it prove that the neighboring rib provides the selected rotational restraint. The analyst must relate this coefficient to the physical joint and the intended sizing policy. The classical dependence of ideal column resistance on stiffness, length and restraint is discussed in [4].

$$P_{cr} = \frac{c\pi^2 (EI)_{min}}{L^2} = \frac{\pi^2 (EI)_{min}}{(KL)^2}, \qquad K = \frac{1}{\sqrt{c}} \tag{37}$$

The code coefficient c is an inverse squared effective-length factor. Default c=1 is the pinned-end Euler idealization.

A path without the required end restraints is not repaired by assigning a large coefficient. For a stringer runout, the metadata can contain a valid shape and complete skin membership while still marking the rib-to-rib restraint model unavailable. Retaining that distinction prevents an unassessed runout from being spread indiscriminately to neighboring properties. A different supported runout idealization would need a separately defined physical model; the existing column equation alone does not supply it.

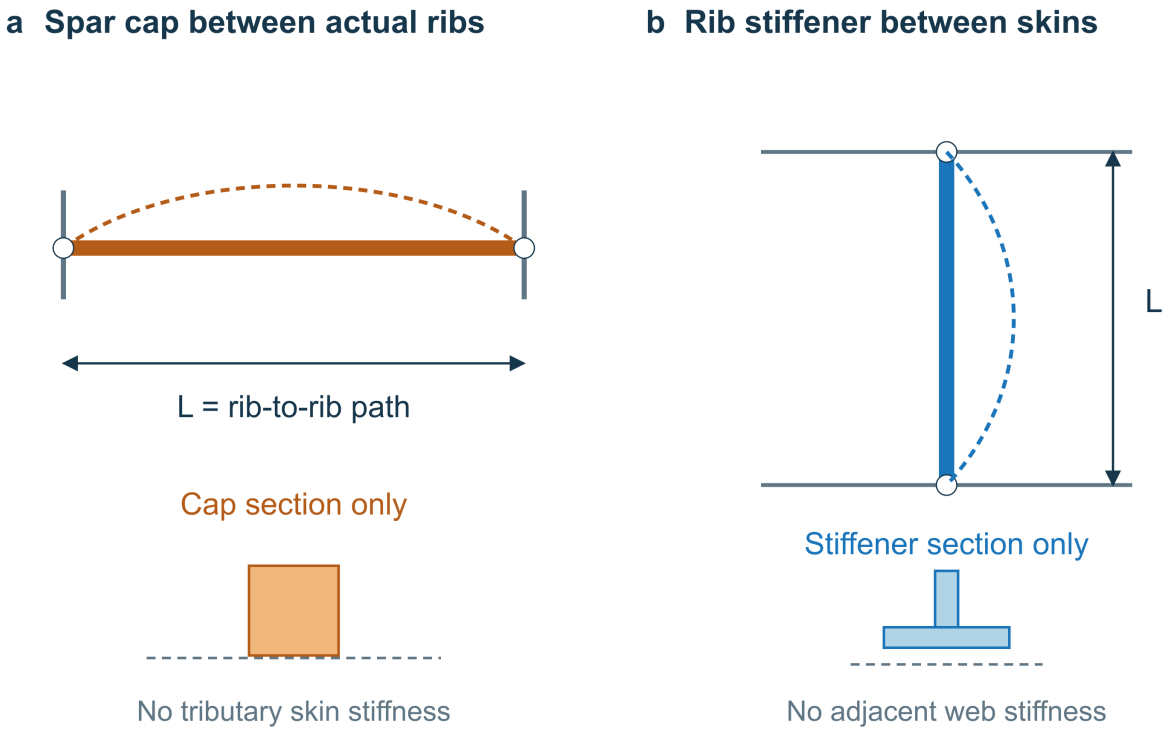


Figure 8: Isolated spar caps and rib stiffeners

Spar-cap and rib-stiffener Euler checks use only the actual isolated member section, unlike the stringer-plus-tributary-skin check. Spar caps use the complete rib-to-rib path; rib stiffeners use the complete skin-to-skin path. The coefficient c represents engineering end restraint and multiplies critical load; the method does not add web or skin stiffness, intermediate elastic bracing, lateral-torsional buckling or crippling.

7.3 Bare spar cap and rib stiffener Euler columns

Spar caps are grouped by actual front or rear cap, upper or lower surface and adjacent rib bay. Their path endpoints must be on the two actual adjacent ribs with modeled rib-web support.

Rib stiffeners are grouped by actual rib and stiffener position. Their path must extend from lower skin to upper skin and have modeled skin support at both endpoints.

These are deliberately bare-member checks. Only the cap or rib-stiffener section and its recovered axial force are included. No adjacent skin or rib-web stiffness or force is credited.

The source uses each current isotropic beam section area and centroidal principal inertias at each bar station. Each station uses the full modeled support-to-support path length. Its local compression and local weak-axis capacity are assessed, and the minimum across all stations/cases governs.

Intermediate elastic bracing, lateral torsional buckling, crippling, local wall buckling, yielding and nonlinear collapse are not represented by this Euler criterion.

The bare-member approach is intentionally different from the stringer-plus-skin model. A cap can be attached to a substantial skin and web in the global FE model, but this local criterion takes no stiffness credit for those neighboring plates. A rib stiffener is similarly assessed from its own section over the complete skin-to-skin height. Those surrounding structures are used to establish support membership, not silently added to the isolated section.

There are therefore two separate questions when reviewing a cap or rib-stiffener RF: whether the member section has enough elastic Euler resistance under its recovered compression, and whether the recorded support condition is a reasonable representation of the real assembly. This criterion does not resolve continuous elastic lateral support or the torsional freedom of an open section. Such effects require another analytical criterion or a suitably modeled FE instability calculation.

Table 6 Column idealizations used by the three Euler families

Criterion	Length and section	Recovered demand
Stringer column	Complete actual rib-to-rib path; stringer plus half the adjacent skin width on each side	Beam axial force plus projected shell membrane forces at one common cut
Spar cap column	Complete actual rib-to-rib cap path; cap section only	Current cap axial force at each bar station
Rib stiffener column	Complete lower-to-upper skin path; stiffener section only	Current stiffener axial force at each bar station

8 Column stiffness and cutwise load extraction

Once membership is established, resistance follows from the actual elastic section. A single geometric second moment is insufficient when the section contains different materials, offsets or coupled laminate strips. Boxer retains the elastic centroid, transverse coupling and weakest principal bending stiffness. It then compares each local capacity with the compression recovered at that same cut.

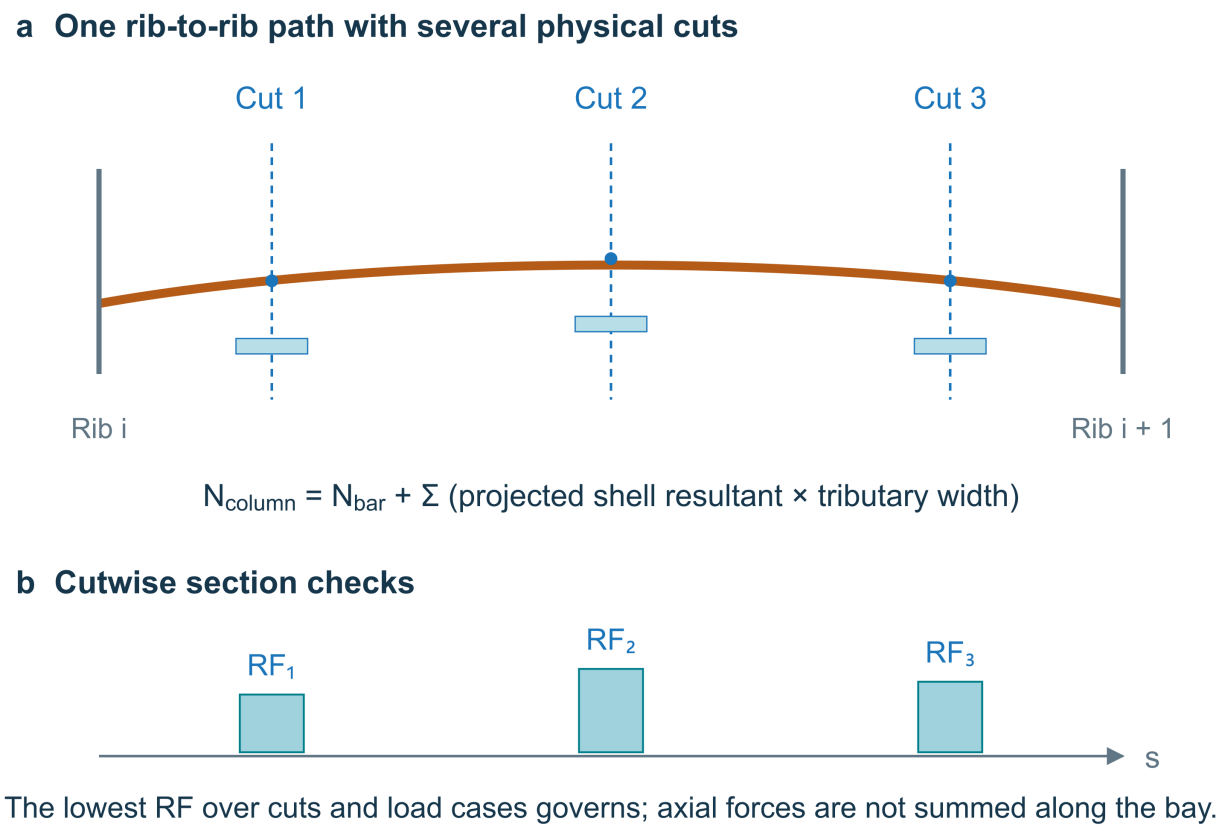


Figure 9: Cutwise Euler demand along a stringer

A stringer-column load is recovered at each actual cross-section cut. The beam axial force and projected skin membrane forces from the same cut are combined. Consecutive axial elements along the rib bay are not added together. Each cut uses its current section stiffness with the full unbraced path length; the lowest reserve factor over all cuts and eligible cases governs.

8.1 Column section stiffness and the weakest bending axis

For a beam T stringer, the outer attached-flange face is on the FE attachment path. Julia splits the actual T into two nonoverlapping rectangles: flange b by tf and free web tw by h-tf. Their eccentric centroids and exact rectangular inertias are retained. A zero-flange rectangular blade is represented by its actual rectangular section.

For isotropic parts the assembled elastic centroid is weighted by EA, not by area alone. The parallel-axis terms are E-weighted. Different isotropic Young moduli can therefore be combined without inventing a common modulus.

For a general shell-strip contribution, a symmetric 3 by 3 axial section law K is retained for extension and its two transverse gradients. The code shifts each contribution to its elastic neutral axis, then assembles all parts with parallel-axis terms. Algebraically this is equivalent to assembling the complete K and eliminating extension once by a Schur complement.

The weakest principal bending stiffness is the smaller eigenvalue of the assembled 2 by 2 bending law. The implementation computes it as determinant divided by the larger eigenvalue to avoid cancellation for slender blades. It rejects a singular or nonpositive section rather than imposing a fictitious stiffness floor.

$$(\bar{y}, \bar{z}) = \frac{\sum_i E_i A_i (y_i, z_i)}{\sum_i E_i A_i} \tag{38}$$

Elastic centroid for uncoupled isotropic parts.

$$(EI)_{yy} = \sum_i \left[E_i I_{yy,i} + E_i A_i (z_i - \overline{z})^2 \right] \tag{39}$$

Bending stiffness about section y, including actual eccentricity.

$$(EI)_{zz} = \sum_i [E_i I_{zz,i} + E_i A_i (y_i - \overline{y})^2] \tag{40}$$

Stiffness about section z includes the elastic parallel-axis contribution.

$$(EI)_{yz} = \sum_i [E_i I_{yz,i} + E_i A_i (y_i - \overline{y})(z_i - \overline{z})] \tag{41}$$

The product-of-inertia coupling is retained.

$$\mathbf{K}_{\text{sec}} = \begin{bmatrix} K_{00} & \mathbf{k}^T \\ \mathbf{k} & \mathbf{K}_{bb} \end{bmatrix}, \qquad \mathbf{H} = \mathbf{K}_{bb} - \frac{\mathbf{k}\mathbf{k}^T}{K_{00}} \tag{42}$$

Complete axial extension is eliminated to obtain the reduced bending law.

$$h_{\max} = \frac{H_{11} + H_{22} + \sqrt{(H_{11} - H_{22})^2 + 4H_{12}^2}}{2} \tag{43}$$

Larger principal eigenvalue of the reduced section bending matrix.

$$(EI)_{\min} = \det \mathbf{H} / h_{\max} \tag{44}$$

Stable weaker principal stiffness avoids subtraction of nearly equal values.

For a homogeneous isotropic assembly the elastic centroid reduces to the ordinary area centroid. With different moduli, a stiff narrow region can influence the neutral axis more than a larger compliant region. The code therefore assembles EA-weighted quantities from the outset. Using a single average E after calculating an area-only inertia would generally give a different result and is not the implemented method.

The off-diagonal product of inertia is retained even when the section is geometrically awkward. Its sign depends on the chosen transverse axes, but the principal stiffness values do not. The smaller principal value governs the free choice of Euler buckling direction. A nearly flat or very slender blade can make subtraction of two similar quadratic eigenvalues inaccurate; calculating determinant divided by the larger eigenvalue avoids that cancellation without imposing an artificial minimum stiffness.

8.2 Laminate strips and oblique section cuts

The full rotated laminate ABD law is formed about the actual shell GRID reference plane, retaining ply angles, material laws, stacking order and shell offset. Isotropic PSHELL is treated as a one-ply physical stack. PCOMP uses its explicit total thickness and z0.

For Euler stiffness, retain axial extension epsilon_x and axial curvature kappa_x. The transverse and shear membrane and curvature variables are condensed together from the full six-variable law. Condensation is not performed independently ply by ply. This is the stated free transverse-relaxation CLT approximation.

Thin-skin curvature variables are scaled by thickness for numerical conditioning during condensation, then restored to dimensional units. For a homogeneous isotropic strip the exact reduced axial law is E times [[t,t o],[t o,t^3/12+t o^2]].

For an oblique shell, p is the magnitude of the column direction projected into the shell tangent plane. The cut thickness is t/p. The axial strain projection produces p^3 A, p^2 B and p D weights in the strip section energy. This is an axial-energy projection approximation for a thin oblique strip.

The implementation preserves laminate extension-bending coupling, but it is not a full anisotropic three-dimensional beam section theory, nor does it add torsional-flexural buckling.

$$\mathbf{C}_{\text{ax}} = \mathbf{C}_{rr} - \mathbf{C}_{rf}\mathbf{C}_{ff}^{-1}\mathbf{C}_{fr}, \quad r = (\epsilon_x, \kappa_x), \quad f = (\epsilon_y, \gamma_{xy}, \kappa_y, \kappa_{xy}) \tag{45}$$

The retained 2 by 2 axial laminate law for the column strip.

$$\mathbf{C}_{\text{ax,iso}} = E\begin{bmatrix} t & to \\ to & t^3/12 + to^2 \end{bmatrix} \tag{46}$$

o is the physical section mid-thickness offset relative to the shell GRID plane.

$$a_x = \mathbf{e}_x \cdot \mathbf{u}, \quad a_y = \mathbf{e}_x \cdot \mathbf{v}, \quad p = \sqrt{a_x^2 + a_y^2} \tag{47}$$

Column direction components in the shell material frame and its tangent-plane projection.

$$\mathbf{K}_A = wp^3 A[\mathbf{v}\mathbf{v}^T + (w^2/12)\mathbf{t}\mathbf{t}^T] \tag{48}$$

Membrane contribution of the reduced axial strip law.

$$\mathbf{K}_B = wp^2 B(\mathbf{v}\mathbf{n}^T + \mathbf{n}\mathbf{v}^T) \tag{49}$$

Extension-bending coupling contribution.

$$\mathbf{K}_D = wpD\mathbf{n}\mathbf{n}^T, \qquad \mathbf{K}_{\text{strip}} = \mathbf{K}_A + \mathbf{K}_B + \mathbf{K}_D \tag{50}$$

Bending contribution and complete strip law. v=[1,y0,z0], t=[0,ty,tz], n=[0,ny,nz].

In the strip matrix equations, w is the actual length of the clipped shell intersection within the section, not the rib-bay length. The vector v locates the strip reference center, t describes the cut tangent, and n gives the transverse direction of the shell normal. The A, B and D symbols here are entries of the reduced two-variable axial laminate law, rather than the complete three-by-three plate matrices. Keeping these roles distinct avoids mixing plate and column constitutive quantities.

Condensation means that the selected transverse generalized forces are set free while axial extension and axial curvature are retained. It is a precise algebraic operation within that idealization, not a statement that the real attached skin is completely unconstrained transversely. Strong constraints imposed by adjoining parts can make another section model more appropriate. The report therefore identifies this freedom rather than treating the condensed law as a universal composite beam stiffness.

For a symmetric laminate with no extension-bending coupling, the reduced law simplifies, but the code does not assume symmetry. It includes the actual stacking order and offset before elimination. A homogeneous isotropic reference-plane shift also creates apparent coupling in the stored law; the assembled neutral-axis elimination removes the reference-plane dependence while preserving the true eccentric distance between the stringer and skin.

8.3 Column compression is recovered at cuts rather than summed along a path

The beam force contribution is the recovered signed forces.axial in newtons. The shell contribution uses the recovered membrane resultants forces.fx, forces.fy and forces.fxy in N/m, projected into the local column direction and integrated over the actual clipped section width.

Contributions are added across the same section only. Consecutive bar or shell elements along the span are not added together. Duplicate beam axial force IDs in a section are rejected.

Native tensile-positive resultants remain signed through the sum; compression demand is the negative signed sum, clipped only at a cancellation-scale floating-point tolerance. There is deliberately no absolute force threshold that could turn genuine small compression into no demand.

Recovered beam bending moments, shell bending moments, von Mises stress peaks and local eccentric load moments are not inserted in this simple axial Euler demand. Eccentric geometry affects section stiffness, but an explicit beam-column interaction or moment magnification criterion is not implemented.

At each cut j and case c, the RF is Pcr,j divided by beta times local compression. The lowest valid value governs. Incomplete section-force or case coverage makes the applicable criterion unassessed.

$$F_{x,e}^{\text{shell}} = \frac{w_e}{p_e}\left(a_{x,e}^2 N_{x,e} + a_{y,e}^2 N_{y,e} + 2a_{x,e}a_{y,e}N_{xy,e}\right) \tag{51}$$

Shell force projection in newtons. The coefficients come from the actual section geometry.

$$F_{x,j} = \sum_{e\in\text{beam cut}} F_{x,e}^{\text{beam}} + \sum_{e\in\text{shell cut}} F_{x,e}^{\text{shell}}, \qquad P_j = \max(0, -F_{x,j}) \tag{52}$$

Compression is positive only after all signed terms on one section have been combined.

$$\text{RF}_{\text{Euler}} = \min_{c,j:P_{cj}>0} \frac{c_E \pi^2 (EI)_{\min,j}}{\beta L^2 P_{cj}} \tag{53}$$

c_E denotes the Euler restraint coefficient, avoiding confusion with case index c.

The signed sum must be formed before deciding whether a cut is in compression. A tensile skin contribution can offset compression in a stringer at the same section, and the opposite can also occur. Summing only the compressive terms would discard that equilibrium interaction; summing the absolute values would be still more misleading. Conversely, adding every axial force from consecutive bars along the path would count the same transmitted force repeatedly.

The force projection uses shell membrane resultants, which carry force per unit cut length. Multiplication by the actual cut width produces newtons. A recovered stress in MPa cannot be inserted into this expression without the appropriate thickness and width integration. This is why the column criterion does not use the stress-report percentile or an element von Mises envelope as its demand.

Using local EI with the full L gives a transparent screening model for a varying member, but it is not the exact Euler solution for an arbitrary EI varying along the path. The code evaluates each station separately and takes the lowest ratio. It does not solve a one-dimensional differential eigenproblem with the actual EI function, eccentric loading, initial curvature and intermediate elastic support. This is an important limit when a favorable local RF is used in a structure with large spanwise changes.

9 Stringer and rib stiffener blade instability

The blade criterion addresses a local instability of an outstand web. It is not an alternative form of the full stringer-column equation. Its resistance depends primarily on web thickness, clear outstand height and isotropic modulus. Rib-bay length is absent because the implemented formula uses the long-plate limit.

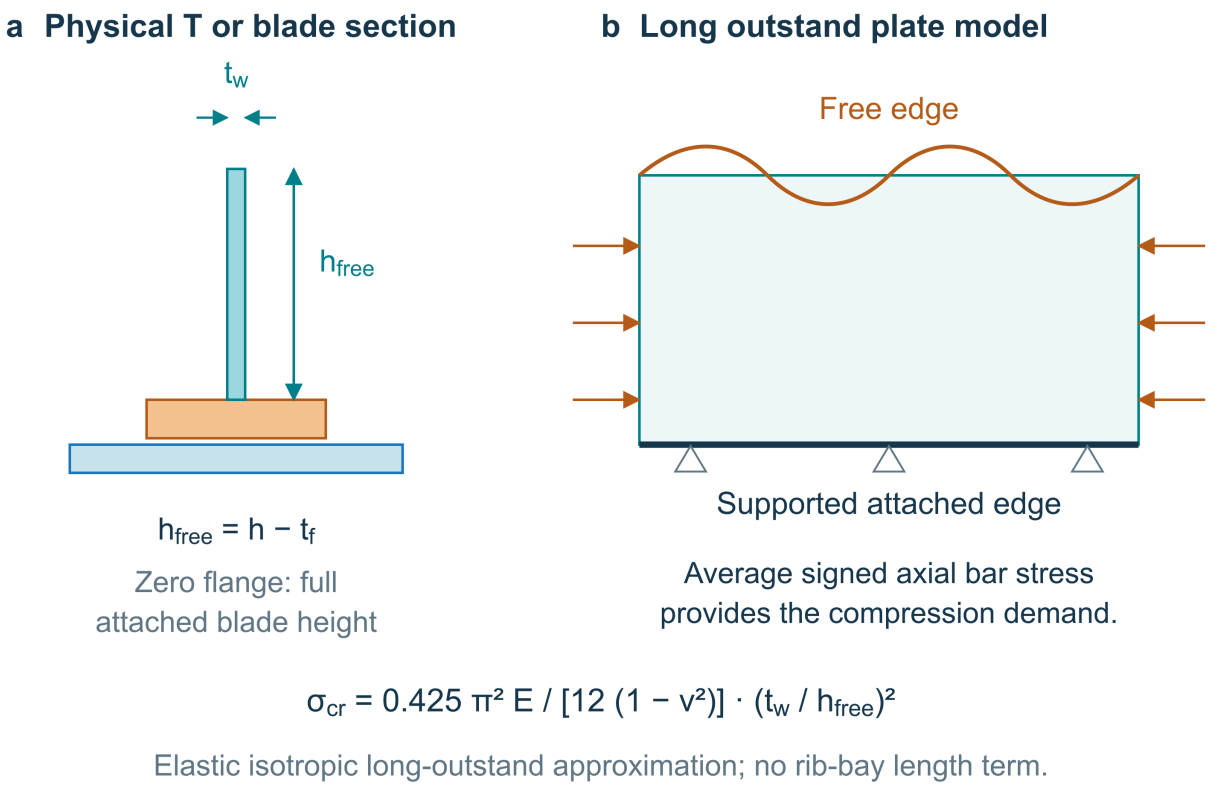


Figure 10: The local free web blade idealization

The free-web check treats the PBARL T web as an isotropic long outstand plate, supported at its attached longitudinal edge and free at the tip. The clear height is the full section height minus flange thickness; a zero-flange rectangular blade uses its full height. The critical stress uses k = 0.425 and the demand is the arithmetic mean of signed axial bar stresses. This local mode is distinct from rib-to-rib Euler buckling.

9.1 Local free blade instability of stringers and rib stiffeners

This mode treats the beam T free web as a long isotropic outstand plate with its attached longitudinal edge supported and its opposite longitudinal edge free. Its clear blade height is h-tf; a zero-flange rectangular beam blade uses its full actual height. Shell-stringer webs are not assessed by this bar-stress blade module.

The coefficient is the hard-coded long-outstand value 0.425. The finite-length stabilizing correction is omitted. Neither rib-to-rib length nor the Euler fixity coefficient appears in this local formula.

For each bar, if endpoint axial stresses are present then both must be finite and their signed average is used. Otherwise a finite signed stress.axial is required. Von Mises or bending stress is never substituted. Stresses are converted from Pa/kPa/MPa/GPa to SI Pa.

Within a physical stringer/rib-stiffener segment, the signed bar stresses are averaged arithmetically with equal element weights, not by length or area. Compensated normalized summation reduces cancellation error. The lowest current blade critical stress among the segment bars is compared with this case mean compression.

Signed averaging can cancel local compression and tension. The code can therefore report no mean compressive demand even while individual elements are compressed. Its maximum local compression is retained diagnostically but does not drive this criterion. This assumption should be reviewed where the axial field varies sharply or the mesh has very unequal bar lengths.

This is elastic web plate buckling, not complete section crippling or a flange-web rotational restraint analysis. Only isotropic MAT1 with positive E and -1<nu<0.5 is eligible.

$$h_{\text{free}} = h - t_f, \qquad \sigma_{\text{cr}} = 0.425\frac{\pi^2 E}{12(1-\nu^2)}\left(\frac{t_w}{h_{\text{free}}}\right)^2 \tag{54}$$

Use clear unsupported blade height rather than full T height. For the zero-flange rectangular blade, use its full height.

$$\sigma_{x,e} = \frac{\sigma_{x,eA} + \sigma_{x,eB}}{2}, \quad \overline{\sigma}_{x,c} = \frac{1}{n_e}\sum_e \sigma_{x,e,c}, \quad \sigma_{\text{comp},c} = \max(0, -\overline{\sigma}_{x,c}) \tag{55}$$

Endpoint averaging when present, then equal-element signed mean over the physical segment.

$$\text{RF}_{\text{blade}} = \min_{c:\sigma_{\text{comp},c}>0}\frac{\min_e \sigma_{\text{cr},e}}{\beta\sigma_{\text{comp},c}} \tag{56}$$

The lowest current blade capacity is combined with the governing mean compressive case.

The long-outstand coefficient is a classical elastic plate idealization with one supported and one free longitudinal edge. NACA TN 3781 provides the background of flat-plate buckling coefficients and support conditions [2]. Boxer implements this particular fixed coefficient, not the handbook as a whole. It does not interpolate a finite-length plate chart or include an inelastic correction.

The visible total T height is not the clear unsupported plate width. The flange occupies part of that height, so the free web starts at the flange junction. Substituting the total height would lower the predicted critical stress relative to the implemented h minus flange thickness definition. For a zero-flange rectangular blade there is no flange region to subtract, and the full height attached along one edge is used.

Equal-element averaging is a meaningful modeling choice rather than a universal integration rule. If a segment is divided into unequal bar lengths, each recovered element stress still receives one equal vote in the current implementation. A different length-weighted or section-force-based averaging rule would be another criterion. The same caution applies when a short but severe compressive region lies alongside a long tensile region: a passing mean-stress check does not eliminate the need to understand the local field.

If section properties vary within a segment, the minimum elastic critical stress among its bars is compared with the case mean compression. This prevents a thin local blade from receiving the capacity of a thicker neighbor, but it does not solve the buckling of a tapered outstand plate. The result remains an elastic local screening check, and its physical scope is retained in the RF details.

The rib-stiffener blade family reuses the same law for an actual isotropic PBARL T stiffener. A spar cap modeled as a solid rectangular section does not acquire a fictitious free-web plate through this reuse. Its implemented cap criterion remains the isolated Euler check plus any applicable material strength assessment.

10 Plate regions materials and membrane fields

A plate RF begins by separating the physical shell region from the rectangle used for its mathematical idealization. The mesh defines which elements, supports, material laws and recovered resultants belong to the region. The rectangle supplies a manageable displacement space. Keeping those two descriptions separate is essential when a wing is swept, tapered, twisted or locally perturbed.

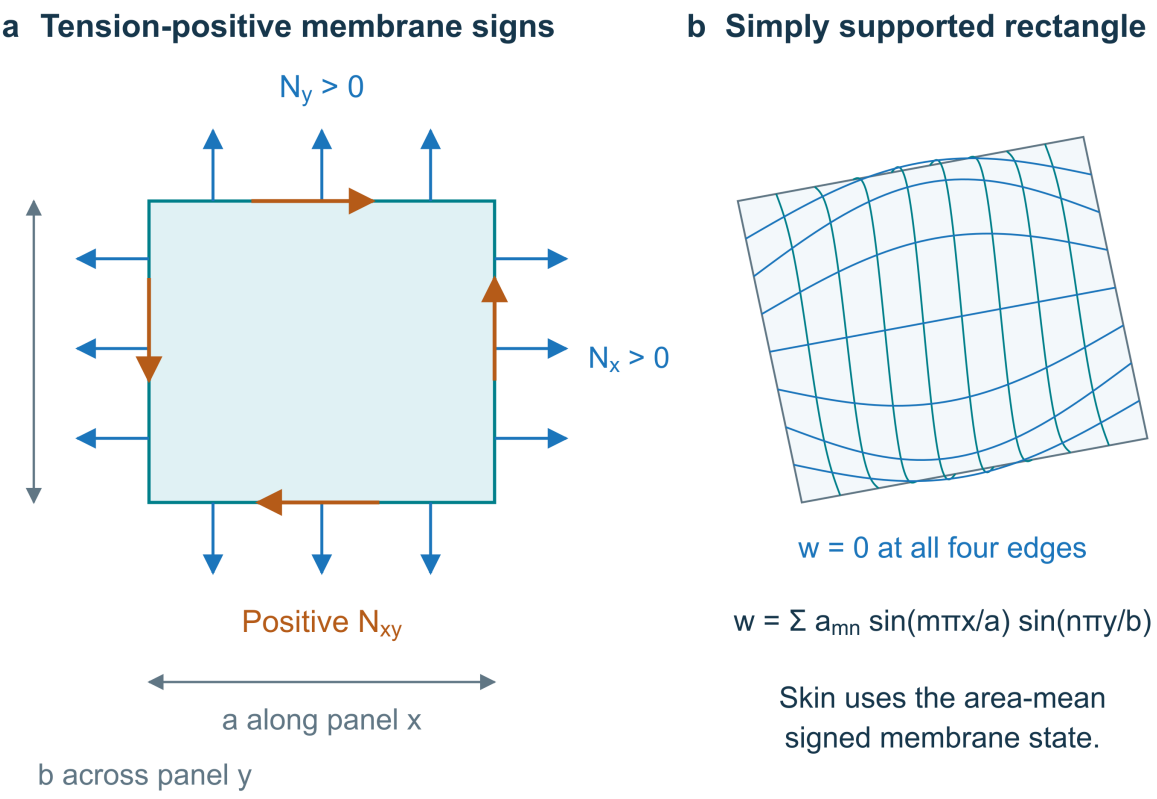


Figure 11: Signed skin loads and the double sine basis

The skin plate receives transformed, signed membrane resultants with tension positive. Its rectangular Ritz idealization uses a double-sine expansion and w = 0 on all four edges. The geometric stiffness uses the compression tensor C = −N, including signed shear. Actual skew, curvature and detailed elastic edge restraint are represented only through the recorded equivalent geometry and engineering fixity coefficient.

10.1 Skin and web regions are physical compartments rather than property IDs

A skin panel is a connected compartment bounded by two adjacent physical ribs and two longitudinal supports, which can be neighboring stringers or a stringer and spar. The topology must form one simple closed four-sided loop with four distinct corners.

The equivalent rectangle length is the mean of the two actual longitudinal boundary arc lengths; its width is the mean of the two actual rib boundary arc lengths. Three-dimensional path lengths are used. Physical area is used separately for averaging; it is not forced to equal the rectangle area.

The panel normal is an area-weighted average of element normals. The x direction is the inboard-to-outboard midpoint direction projected into the panel plane; y completes the right-handed frame. Each shell material frame is projected/normalized into that common frame.

Nonmanifold edges, holes, unsupported boundaries, terminating internal supports, non-four-sided runout-merging compartments, folds and inconsistent normals are not silently converted into ordinary rectangles. Their criteria remain unassessed with explanatory metadata.

Spar webs are compartmentalized between actual rib support paths and upper/lower skin paths. Rib webs are compartmentalized between front/rear spars or actual vertical rib stiffeners and upper/lower skin paths. Each equivalent web rectangle has the same mean-opposite-boundary dimension convention.

The web equivalent-coordinate map averages the normalized arc coordinates on the two opposite skin boundaries and uses modeled through-height fraction. The browser presently requires each mapped face to be a four-corner axis-aligned rectangle in this coordinate map, with complete nonoverlapping tiling. Triangular or unsupported mapped cells are not silently approximated by rectangles.

$$a = \frac{L_{\text{left}} + L_{\text{right}}}{2}, \qquad b = \frac{L_{\text{inboard}} + L_{\text{outboard}}}{2} \tag{57}$$

Skin equivalent dimensions from actual 3D support paths.

An equivalent rectangle is not an isoparametric reconstruction of every physical shell surface. Its dimensions preserve the specified average boundary path lengths and its frame follows the recorded mean geometry, but it does not retain every skew angle, local curvature or variable support stiffness in the buckling operator. The metadata records skew and normal deviation for review; these geometric diagnostics should not be confused with additional stiffness terms in the analytical plate equation.

Property regions can cross the boundaries of a physical assessment, and one physical panel can contain several properties. Neither coincidence nor mismatch is an error by itself. The important question is whether every actual member element has a verified place in the compartment and whether the intended supports close it. A shared RF describes that complete compartment, even when it is displayed on several of its property rows.

10.2 Laminate bending rigidity for the plate assessments

Each actual element property is converted into a rotated full CLT ABD matrix in the common panel direction. Ply interfaces use the real reference-plane offset. The effective bending rigidity is D*=D-B^T A^-1 B, calculated using a Cholesky solve for A and checked for positive definiteness.

D* follows free membrane relaxation of incremental bending. It is independent of a mere shift of reference plane when the full offset coupling is retained, but for an unsymmetric laminate it is still an equivalent-plate approximation. It is not the full coupled membrane-bending buckling boundary-value problem.

The actual element D* matrices are area-weighted to one constant effective bending matrix for the physical panel. Variable thickness, material and layup are therefore homogenized in bending, even in the spatial web method.

The local plate model is thin Kirchhoff CLT. It omits transverse shear deformation. A sandwich can have a positive D* and still be outside a reliable thin-plate approximation if core shear deformation or face/core instabilities dominate.

$$\mathbf{A} = \sum_k \overline{\mathbf{Q}}_k (z_k^1 - z_{k-1}^1) \tag{58}$$

Ply integration of A, units N/m. Ply positions include the actual reference-plane offset.

$$\mathbf{B} = \frac{1}{2} \sum_k \overline{\mathbf{Q}}_k (z_k^2 - z_{k-1}^2) \tag{59}$$

Ply integration of B, units N. Ply positions include the actual reference-plane offset.

$$\mathbf{D} = \frac{1}{3} \sum_k \overline{\mathbf{Q}}_k (z_k^3 - z_{k-1}^3) \tag{60}$$

Ply integration of D, units N m. Ply positions include the actual reference-plane offset.

$$\mathbf{D}^* = \mathbf{D} - \mathbf{B}^T \mathbf{A}^{-1} \mathbf{B} \tag{61}$$

Free membrane relaxation of the incremental bending law.

$$\overline{\mathbf{D}} = \frac{\sum_e A_e \mathbf{D}_e^*}{\sum_e A_e} \tag{62}$$

Constant area-weighted bending rigidity used by the equivalent rectangle.

$$D_0 = \frac{Et^3}{12(1-\nu^2)}, \qquad \mathbf{D}_{\text{iso}} = D_0 \begin{bmatrix} 1 & \nu & 0 \\ \nu & 1 & 0 \\ 0 & 0 & (1-\nu)/2 \end{bmatrix} \tag{63}$$

Isotropic thin-plate special case, using engineering curvature convention.

The A inverse operation in the relaxed bending law is implemented as a factorization and solve, not by explicitly forming an unstable inverse matrix. Symmetry is restored against roundoff and the resulting D star must be positive definite. A laminate with unsupported or nonphysical material data is rejected at this stage rather than passed into an eigenproblem with an arbitrary regularization.

Area averaging the bending law is another idealization. It gives one constant bending tensor even where the original panel has local thickness changes, mixed materials or ply transitions. The spatial web method improves the representation of the preload field, but it still uses this homogenized bending tensor. A small thick patch does not appear as an independently placed stiffener in the analytical plate stiffness.

Classical laminate ABD definitions and their dependence on stacking order are described in NASA RP 1351 [1]. The present use of a relaxed D star adds an explicit assessment assumption. In particular, a thick sandwich may need transverse-shear compliance or local face and core criteria that this thin-plate model does not contain. A positive algebraic bending matrix alone is not a check that all those omitted mechanisms are harmless.

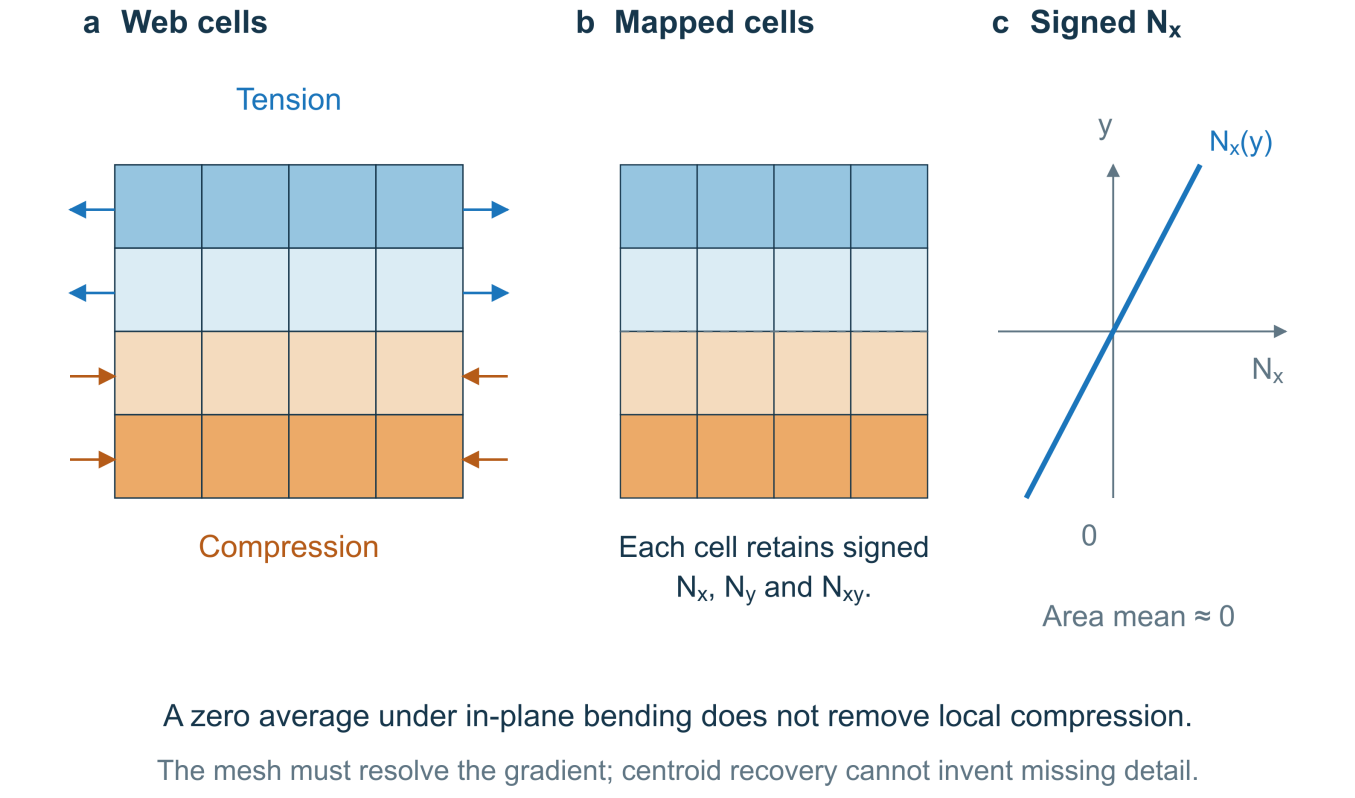


Figure 12: Spatial web loading with zero mean compression

Spar and rib web buckling retains a piecewise-constant signed membrane field over mapped element cells. In-plane bending can produce tension and compression with nearly zero mean; the code still integrates the local destabilizing field. Displayed mean stresses are context only. A coarse mesh with no through-height recovery cannot reveal a sub-element gradient that was never exported.

10.3 Signed membrane recovery and the difference between skin and web loading

Recovered shell membrane resultants are tensile-positive. The 3-vector [fx,fy,fxy] is rotated from each result material frame to the common panel frame. The buckling solver changes the sign of all three components so its compression tensor is positive in compression.

For SKIN panels, the transformed membrane resultants are area-averaged to a uniform tensor before the buckling eigenproblem. The displayed stresses are separately area-averaged element N/t. If thickness varies, mean(N/t) is not mean(N)/mean(t). Stress-percentile trimming never enters this mode.

For SPAR and RIB WEB panels, the displayed means are contextual only. Each element tensor is retained as a piecewise-constant field on its recorded equivalent-rectangle cell. Compression and tension gradients and signed shear are integrated into the geometric matrix; opposing tensile and compressive regions are not erased by a zero panel mean.

The web result field is only as detailed as the actual recovered mesh. One element through the web height does not reveal an intra-element bending stress gradient absent from its centroid membrane resultant. The method does not fabricate that gradient from extreme-fiber stress.

All required actual elements in each selected physical case must have finite signed membrane resultants and verified compatible units. Missing data cannot be substituted with von Mises stress or an assumed zero.

$$\mathbf{T}_N(c,s) = \begin{bmatrix} c^2 & s^2 & 2cs \\ s^2 & c^2 & -2cs \\ -cs & cs & c^2 - s^2 \end{bmatrix}, \qquad \mathbf{N}_e^p = \mathbf{T}_N \mathbf{N}_e^m \tag{64}$$

c and s locate the panel x projection in the shell material frame.

$$\overline{\mathbf{N}} = \frac{\sum_e A_e \mathbf{N}_e^p}{\sum_e A_e}, \qquad \mathbf{C}_{\text{skin}} = -\overline{\mathbf{N}} \tag{65}$$

Uniform skin membrane demand is the negative signed area mean.

$$\overline{\boldsymbol{\sigma}} = \frac{\sum_e A_e (\mathbf{N}_e^p / t_e)}{\sum_e A_e} \tag{66}$$

The contextual mean stress differs from mean membrane resultant divided by mean thickness.

$$\mathbf{C}_{\text{web}}(x,y) = -\mathbf{N}_e^p, \qquad (x,y) \in \Omega_e \tag{67}$$

Piecewise signed web demand on each mapped cell.

The difference between skin and web demand is especially important under in-plane bending. A skin compartment with tensile and compressive subregions can have a small averaged compression tensor; its present uniform-field model uses that average. A web compartment retains those subregions individually. It can therefore predict an instability in the compressed region even when the displayed mean membrane stress is zero.

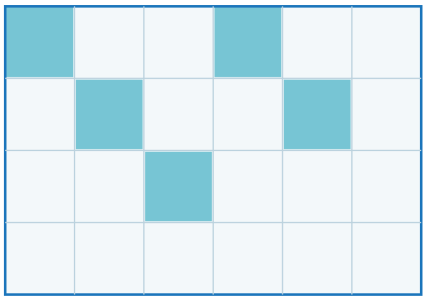
This does not make the web calculation independent of FE mesh resolution. The element resultants are treated as constant on their mapped cells. If a true linear stress gradient is represented by several cells, its sign changes can enter the geometric energy. If it is represented by one centroid sample, a zero centroid resultant contains no information about the tensile and compressive extremes. The RF postprocessor cannot reconstruct information that the result payload does not provide.

Mean stress, membrane resultant and surface stress are consequently three different quantities. Mean N divided by a mean thickness is generally not the same as the recorded area mean of N divided by each element thickness. Neither is a top-face bending stress. The report and tables retain these distinctions so a stress contour is not accidentally interpreted as the field used by the plate eigenproblem.

11 Rayleigh Ritz formulation and eigenvalue solution

The plate calculation is an energy-based Rayleigh Ritz eigenproblem. It retains biaxial compression, tensile stabilization, signed shear and anisotropic bending coupling in one operator. This is more specific than selecting a compression coefficient and adding an empirical shear interaction formula. The following equations describe the actual idealized problem solved by the local plate modules.

a Compatible sine modes



$$w = \Sigma\, a_{mn} \sin(m\pi x/a) \sin(n\pi y/b)$$

Exact curvature and gradient energy integrals

b Symmetric eigenproblem

$$c\, K_b\, a = RF_{limit}\, G\, a$$

Bending stiffness: positive definite
Geometric term: signed load field

Nested basis refinements

Two successive RF changes at or below 0.5% → converged

Unresolved source basis

Select at most 100 physical modes and re-solve → labelled approximation

$$RF_{policy} = RF_{limit} / \text{buckling policy factor}$$

A finite truncated-basis result can overestimate capacity; it is not a convergence certificate.

Figure 13: Ritz convergence and one hundred term approximation

Plate buckling uses a double-sine Ritz displacement space with analytic energy integrals and a symmetric generalized buckling problem. Nested refinements require two successive relative RF changes no larger than 0.5%. If the source basis remains unconverged, at most 100 physical sine terms are selected and re-solved; a finite result is explicitly marked as an approximation that may overestimate capacity. The policy factor rescales the RF independently of stiffness or physical recovered loads.

11.1 Rayleigh Ritz plate formulation

The admissible transverse displacement is expanded in double sine functions. The basis enforces zero transverse displacement on the four rectangle edges. The intended idealization is simply supported. Natural bending moment conditions are represented variationally; for coupled anisotropic D16/D26 they are not necessarily satisfied pointwise by each finite sine basis function.

The code includes the complete symmetric effective bending law, including D16/D26. The geometric energy contains signed biaxial compression and shear in the original common panel axes. It does not replace the coupled problem with a scalar compression/shear interaction equation or drop tensile stabilization.

Let kappa=[-w,xx,-w,yy,-2w,xy]. The bending matrix is the integral of curvature products; the geometric matrix is the integral of in-plane tensor work on transverse gradients. With compression-positive C, the instability condition is c Kb q = lambda G q. The buckling-policy RF is lambda/beta.

For uniform skin loads, the common mode norm ab/4 is divided out. Diagonal bending and normal geometric terms are analytic. D16/D26 and shear create off-diagonal coupling only when both longitudinal and transverse parity change. The invariant m+n parity sectors are both solved so the antisymmetric family cannot be missed.

For spatial web loads, exact sine/cosine product integrals are evaluated over every mapped rectangle cell. Spatial variation generally destroys the uniform-load parity separation, so the complete matrix is solved.

$$w(x,y) = \sum_{m=1}^{M}\sum_{n=1}^{N} q_{mn}\sin\left(\frac{m\pi x}{a}\right)\sin\left(\frac{n\pi y}{b}\right) \tag{68}$$

Admissible double-sine Ritz displacement field. Define the modal wavenumbers alpha_m=m*pi/a and k_y,n=n*pi/b; m and n are positive integers.

$$U_b = \frac{1}{2}\int_\Omega \boldsymbol{\kappa}^T\overline{\mathbf{D}}\boldsymbol{\kappa}\,dA = \frac{1}{2}\mathbf{q}^T\mathbf{K}_b\mathbf{q} \tag{69}$$

Bending strain energy; engineering twist curvature is -2 w,xy.

$$W_g = \frac{1}{2}\int_\Omega\left(C_x w_{,x}^2 + 2C_{xy}w_{,x}w_{,y} + C_y w_{,y}^2\right)dA = \frac{1}{2}\mathbf{q}^T\mathbf{G}\mathbf{q} \tag{70}$$

Destabilizing geometric work for the complete signed membrane tensor.

$$\left(c_p\mathbf{K}_b - \lambda\mathbf{G}\right)\mathbf{q} = \mathbf{0}, \qquad \mathrm{RF} = \frac{\lambda}{\beta} \tag{71}$$

c_p is the engineering plate fixity coefficient. It does not change the sine basis.

$$\frac{2}{a}\int_0^a \sin\left(\frac{m\pi x}{a}\right)\cos\left(\frac{p\pi x}{a}\right)dx = \begin{cases} 0, & m+p \text{ even} \\ \dfrac{4m}{\pi(m^2-p^2)}, & m+p \text{ odd} \end{cases} \tag{72}$$

The implemented exact normalized sine-cosine integral underlying uniform shear and anisotropic coupling.

The sign convention must be read with the energy equation. In this chapter, the local matrix G is assembled from a compression-positive tensor C. Positive geometric work destabilizes the plate. Native whole-model SOL105 uses its own geometric-stiffness sign convention and is described separately in the report. The two symbols should not be exchanged solely because both are sometimes called geometric stiffness.

A double-sine space allows the plate to move out of plane while vanishing at every edge. It does not set the normal slope to zero, so it is not a clamped basis. For an isotropic or specially orthotropic plate, familiar simply supported solutions can be represented directly. With extension-relaxed anisotropic bending and D16 or D26 coupling, the finite basis satisfies the essential zero-displacement conditions, while natural edge moments arise through the variational statement rather than being imposed pointwise on each individual trial function.

The energy viewpoint also clarifies why shear and tension cannot be replaced by their absolute values. Tension contributes a stabilizing sign in the geometric form. Shear can orient destabilizing gradients obliquely, and changing its sign can change the mode pattern. The code preserves the full signed tensor in the common panel frame. Classical plate energy and Rayleigh Ritz derivations are discussed in [3].

11.2 Exact cell integration for spatial web fields

On normalized coordinates u=x/a and v=y/b, each mapped shell cell is [u0,u1] by [v0,v1] with constant signed C. Products of sine and cosine are integrated through product-to-sum identities, not sampled at a grid of quadrature points.

The code forms cc, ss, cs and sc integral tables for every distinct cell interval. Multiplying the x and y tables gives the exact product integral for that cell. A factor 4 restores the common ab/4 mode normalization convention used in the bending matrix.

The mapping must tile the entire unit rectangle without holes or overlaps. This map and the constant averaged bending rigidity are idealizations of the physical web; exact algebraic integration does not make the geometric idealization exact.

$$I_c(k;l,h)=\begin{cases} h-l, k=0 \\ \dfrac{\sin(k\pi h)-\sin(k\pi l)}{k\pi}, k\neq 0 \end{cases} \tag{73}$$

Exact normalized interval integral of the cosine primitive.

$$I_s(k;l,h)=\begin{cases} 0, k=0 \\ \dfrac{\cos(k\pi l)-\cos(k\pi h)}{k\pi}, k\neq 0 \end{cases} \tag{74}$$

Exact normalized interval integral of the sine primitive.

$$\int_l^h \cos(m\pi u)\cos(p\pi u)du=\frac{I_c(m-p;l,h)+I_c(m+p;l,h)}{2} \tag{75}$$

Cosine-cosine cell integral; analogous product-to-sum formulas give sine-sine and mixed terms.

$$(G)_{ij}=\sum_e \int_{\Omega_e}\left[C_{x,e}\phi_{i,x}\phi_{j,x}+C_{y,e}\phi_{i,y}\phi_{j,y}+C_{xy,e}(\phi_{i,x}\phi_{j,y}+\phi_{i,y}\phi_{j,x})\right]dA \tag{76}$$

The web geometric matrix retains actual signed piecewise membrane fields.

The exact interval products make the spatial integration deterministic for the assumed cell field. There is no choice of a low quadrature order that might miss a narrow compressive cell. However, the word exact applies to integrating the adopted Fourier functions against the adopted piecewise-constant field. It does not remove the errors introduced by an equivalent rectangular map, constant averaged bending law, centroid recovery or a truncated displacement space.

For a truly uniform cell field, the web module calls the uniform plate solver. This is an algebraic simplification of the same energy problem, not a switch to a different safety coefficient. When the cells are nonuniform, their spatial variation can couple mode families that were independent for a uniform rectangle, so the full spatial matrix is retained.

11.3 The numerical eigenvalue algorithm and signed tensile stabilization

The bending matrix must be symmetric positive definite. The code factors Kb=L L^T and constructs H=L^-1 G L^-T by triangular solves, then explicitly symmetrizes roundoff. This is a symmetric congruence, not an unsymmetric inverse product.

The largest algebraic eigenvalue mu of H gives the unit-fixity positive multiplier 1/mu. The engineering plate coefficient then gives lambda=c_p/mu, and the buckling-policy RF is c_p/(beta*mu). Tensile stabilization can make H indefinite with large negative eigenvalues. A magnitude-based power method could select the wrong sign; it is not used.

The implementation scales H, performs symmetric Householder tridiagonal reduction, obtains a Gershgorin eigenvalue bracket and uses a Sturm count with bisection to isolate the largest algebraic eigenvalue. Up to 100 bisection iterations and floating-point relative bracket criteria are used.

Where a parent eigenvector is needed for truncation, shifted tridiagonal inverse iteration is performed for ten iterations and the Householder transformations are reversed. Physical Fourier coefficients are recovered with L^-T before term ranking.

A positive eigenvalue at the estimated numerical resolution is not accepted as a finite RF. In the uniform skin kernel the resolution is 128*n*machine_epsilon times a row-sum contribution bound. Weak compression is not changed into the RF 99 marker merely because the eigenvalue is unresolved.

When D16=D26=0, the sine bending matrix is exactly diagonal. The congruence then uses square roots of the diagonal rather than a cubic Cholesky factorization. Both parity sectors and signed geometric terms remain present.

$$\mathbf{K}_b = \mathbf{L}\mathbf{L}^T, \qquad \mathbf{H} = \mathbf{L}^{-1}\mathbf{G}\mathbf{L}^{-T} \tag{77}$$

Symmetric normalized geometric operator.

$$\lambda_{\mathrm{cr}} = \frac{c_p}{\mu_{\max}}, \qquad \mathrm{RF} = \frac{c_p}{\beta\mu_{\max}}, \qquad \mu_{\max} > 0 \tag{78}$$

The largest algebraic eigenvalue mu_max of H controls; the plate coefficient and buckling policy are applied independently.

11.4 Numerical normalization and the units of diagnostic eigenvalues

The energy-integral matrices in the preceding derivation multiply physical transverse amplitudes measured in metres. Their units are N/m, so q transpose Kb q and q transpose G q are energies in N m. The sine kernel divides both matrices by the common modal area A0=ab/4. This common scaling cancels from the generalized eigenvalue. Stored bending coefficients such as D11 alpha^4 consequently have units N/m cubed, rather than N/m. The symbols used in the energy derivation and the literal stored arrays must therefore be distinguished when checking dimensions.

$$A_0 = ab/4, \qquad \tilde{\mathbf{K}}_b = \mathbf{K}_b/A_0, \qquad \tilde{\mathbf{G}} = \mathbf{G}/A_0 \tag{79}$$

Common mode-area normalization changes matrix units, but not the buckling factor.

The browser also separates demand magnitude from its signed direction so that proportional cases can reuse a factorization and normalized solution. For a skin panel, s_N is the largest absolute component of its averaged compression tensor. For a spatial web, s_N is the largest absolute component over every mapped cell. Each signed tensor is divided by this same positive s_N; tensile components and the sign of shear are retained. The zero-demand path is handled separately before this normalization.

$$s_N = \max_{e,i}|C_{e,i}|, \qquad \hat{\mathbf{C}}_e = \mathbf{C}_e/s_N \tag{80}$$

A uniform skin uses one tensor; a spatial web uses all cell tensors. In SI, s_N has units N/m.

Let mu-hat be the largest algebraic eigenvalue computed with the demand-normalized geometric matrix. It has units m/N when the normalized demand direction is dimensionless. The largest eigenvalue mu associated with the actual physical demand is s_N times mu-hat. The raw largestEigenvalue diagnostic is therefore not generally the reciprocal of the returned case RF. The plate restraint coefficient is applied after the numerical solution, and the independent buckling policy divides that result.

$$\mu_{\max} = s_N\hat{\mu}_{\max}, \qquad \mathrm{RF} = \frac{c_p}{\beta s_N\hat{\mu}_{\max}} \tag{81}$$

The policy factor beta is also denoted f_b. Equivalently RF=c_p/(f_b s_N mu-hat). The physical-G equation RF=c_p/(beta mu) remains unchanged.

One diagnostic detail in the audited implementation needs care. The skin solvePlate path rescales both its final factor and refinement-history factors by 1/s_N. The spatial-web solveSpatialPlate path rescales its final factor by 1/s_N, but retains normalized trial factors in its history; assessWeb then applies the restraint coefficient and policy to those history entries without the missing 1/s_N. The final spatial-web RF is correctly rescaled, and the relative changes used to judge convergence are unchanged because a common positive scale cancels. However, the web history numbers should currently be read as normalized internal trial factors rather than literal final case RFs. This is an identified diagnostic scaling discrepancy, not a change to the final capacity calculation. No application code was changed for this report.

The manufactured zero-mean spatial-web example illustrates the distinction: s_N is 10000 N/m, the final unit-fixity, unit-policy RF is 15.73996870, and the last stored normalized trial factor is 157399.6870. Their ratio is the load normalization s_N. A normalized history value must not be compared directly with the RF acceptance threshold of one.

A largest-magnitude eigenvalue algorithm would be unsuitable for this task. An indefinite signed load field can create a large negative eigenvalue associated with tensile stabilization while the physically relevant destabilizing eigenvalue is smaller and positive. The implementation therefore seeks the largest algebraic value of the symmetric normalized operator. Both the sign and the numerical resolution of that value matter before taking its reciprocal.

The eigenvalue tolerance within the finite matrix and the refinement tolerance of the Ritz space measure different things. A precisely isolated finite-matrix eigenvalue can still belong to a displacement space that is too small. Conversely, failure to resolve a very weak positive eigenvalue is not proof of zero compressive demand. These outcomes have separate report states instead of being folded into a universal passing RF.

The factorization and normalized geometric matrices can be reused for cases having the same geometry and bending law. Demand magnitude is separated from its normalized signed direction, so proportional load cases can reuse the same finite-basis solution and rescale the resulting factor. Such caching changes computational work, not the underlying load policy, signed field or physical coverage requirements.

12 Convergence approximations and assessment status

The code distinguishes a converged refinement sequence, an explicitly truncated approximation, a certified non-governing unresolved case and a genuinely unassessed criterion. These are different mathematical statements. A single numerical RF without its status cannot communicate which statement was established.

12.1 Convergence and the uncoupled exact integer mode route

For uniform uncoupled orthotropic or isotropic bending with zero shear demand, sine modes are uncoupled. The code searches integer m,n values for the minimum positive modal quotient. It starts with low modes and axial analytical candidates, derives a conservative high-mode bound, and enumerates every remaining integer candidate within the bound. It does not assume m=n=1.

For coupled skin problems, odd refinement parameters q=3,5,... allocate longitudinal and transverse terms according to geometry and orthotropic stiffness aspect. A fast family is limited to 256 terms. If needed, a load-direction family enriches short-wave directions up to the 1024-term interactive limit.

Two consecutive relative changes must be at most 0.005 within the same nested family and both modal dimensions must increase. A newly changed aspect allocation starts a fresh stability sequence. The proposed converged value must not be worse than the lowest known prior value beyond floating-point tolerance.

Spatial web problems use a geometry-based family q=3,5,...,19 with default maximum 256 terms. Two consecutive changes at most 0.005 are required. Uniform cell fields dispatch to the uniform plate kernel.

The 0.5 percent condition is an empirical refinement criterion for this model and mode allocation. It is not a rigorous bound on continuum error, nor a comparison with a different FE solver. Large mode counts can still leave a short-wave or oblique state unresolved.

$$\lambda_{mn} = \frac{D_{11}\alpha_m^4 + 2(D_{12} + 2D_{66})\alpha_m^2 k_{y,n}^2 + D_{22}k_{y,n}^4}{C_x\alpha_m^2 + C_y k_{y,n}^2}, \qquad C_x\alpha_m^2 + C_y k_{y,n}^2 > 0 \tag{82}$$

Uncoupled uniform normal-load modal quotient.

$$\delta_j = \frac{|\lambda_{j-1} - \lambda_j|}{\lambda_j}, \qquad \delta_{j-1} \le 0.005, \quad \delta_j \le 0.005 \tag{83}$$

Two successive acceptable nested refinements are required.

$$\rho = \frac{a}{b}\left(\frac{D_{22}}{D_{11}}\right)^{1/4} \tag{84}$$

Geometry and orthotropic stiffness based initial modal aspect allocation.

An uncoupled normal-load rectangle can have many possible longitudinal half-wave counts, especially when its length is large compared with its width. The exact integer search therefore checks more than the first sine shape. Its high-mode bound proves that omitted integer candidates cannot lower the best value within this uncoupled equivalent-plate model. That is stronger than assuming that a plausible-looking first mode must govern.

Coupled problems need nested enrichment because different trial functions interact through shear or bending coupling. Increasing only one modal direction can make a sequence appear stable while an unresolved short-wave pattern remains in the other. The skin solver requires growth in both dimensions within the same family. A change to load-direction enrichment restarts the successive-change test, rather than comparing incompatible basis allocations as if they were one nested sequence.

The size limits are interactive computational limits, not geometric limits imposed by continuum mechanics. They keep a browser assessment manageable. An extreme aspect ratio or weak oblique compression under strong transverse tension may still need more modes than the current family can provide. The fallback and status fields exist to expose that limitation; they do not establish that the continuum problem has become undefined.

12.2 Explicit one hundred term approximations after nonconvergence

The current code implements the user-requested fallback: when a full refinement family has not converged, use the parent basis with the lowest finite positive factor, obtain its eigenvector, rank physical sine functions by qi^2*(Kb)ii, retain at most 100 terms, and re-solve the selected signed energy matrices.

For the uniform skin parity decomposition, the budget is apportioned between both parity sectors. Physical sine functions are selected, not arbitrary mixtures in Cholesky coordinates. Webs use the complete spatial geometric matrix when selecting and re-solving.

A finite positive result is marked approximation=true and converged=false. Its selected modes, source basis dimensions, basis size and reason are retained. It can participate in the property minimum RF, but is never described as a converged plate solution.

A restricted Ritz subspace can overestimate capacity. Reducing the basis does not cure physical or numerical nonconvergence, and a displayed RF above 1 from this fallback is not evidence that the complete plate problem is verified safe.

If there is no valid parent basis or no resolved positive eigenvalue in the truncated basis, the fallback cannot produce a finite RF; the applicable criterion remains unassessed unless a separate non-governing certificate is possible.

$$r_i = q_i^2 (K_b)_{ii}, \qquad \mathcal{J} = \mathrm{Top}_{100}\{r_i\} \tag{85}$$

Implemented physical Fourier term ranking; both parity sectors receive a budget in the uniform problem.

$$\left(\mathbf{K}_{b,\mathcal{J}} - \lambda_{\mathcal{J}} \mathbf{G}_{\mathcal{J}}\right)\mathbf{q}_{\mathcal{J}} = \mathbf{0} \tag{86}$$

The selected subspace is re-solved, rather than copying the parent eigenvalue. This is the unit-fixity pencil; c_p and beta are applied afterward.

The selected terms are the original physical sine functions. The parent eigenvector is first transformed out of normalized bending coordinates before their contributions are ranked. Keeping arbitrary Cholesky-coordinate components would define another reduced space and would no longer have the stated physical Fourier interpretation. Re-solving the selected bending and signed geometric matrices preserves a clear mathematical meaning for the approximation.

There is an important direction to the approximation risk. Restricting an admissible Ritz space can exclude the actual most destabilizing shape and thereby raise the estimated critical load. A smaller one-hundred-term space is therefore not automatically conservative merely because it uses fewer functions. The interface marks the result as an approximation and keeps the unconverged source history; the numerical value is not promoted to a convergence certificate.

12.3 Conservative lower bounds certify only non governing unresolved cases

When a finite truncated value is unavailable, the code can sometimes show that an unresolved case cannot govern another fully converged selected case. It combines a positive Gershgorin lower bound d on bending rigidity with an upper bound n on the largest positive principal membrane resultant and the rectangular Poincare inequality.

For skin, n comes from the averaged compression-positive tensor. For webs, the maximum positive principal value over all cells is used. If the Gershgorin lower bound is nonpositive, no certificate is issued even if the true matrix is positive definite.

The lower bound is multiplied by c and divided by beta. For a skin displayed bound, downward five-significant-figure quantization keeps formatting conservative. A certificate is accepted only if its bound exceeds a separately converged governing case by the stated margin.

A certificate is not an estimated missing eigenvalue. The unresolved case remains labelled unresolved and retains its lower bound and explanation. Missing membrane recovery cannot be certified by this route.

$$d = \min_i \left(D_{ii} - \sum_{j \neq i} |D_{ij}| \right) > 0 \tag{87}$$

Gershgorin lower bound on the smallest eigenvalue of the effective bending matrix, with conservative roundoff padding in code.

$$n = \max_e \lambda_{\max}\left(\begin{bmatrix} C_{x,e} & C_{xy,e} \\ C_{xy,e} & C_{y,e} \end{bmatrix} \right) > 0 \tag{88}$$

Use a single tensor for skin or the maximum over spatial web cells.

$$\mathrm{RF}_{\mathrm{lower}} = \frac{c_p}{\beta} \frac{d\pi^2 (a^{-2} + b^{-2})}{n} \tag{89}$$

Conservative lower bound on every positive buckling factor of this equivalent plate.

A lower-bound certificate serves a different purpose from the reduced-space RF. It can establish that a difficult case is unable to govern an already converged case without claiming to know the difficult case eigenvalue. The comparison is meaningful only after complete membrane recovery and positive material geometry checks. It cannot repair a missing cell, an unsupported boundary or an unknown load case.

If the bound is weaker than the known governing RF, the code has not proved the unresolved case harmless. The combined assessment must then retain an unavailable state unless another explicitly allowed finite approximation is obtained. Similarly, a failed positive Gershgorin estimate does not mean the material matrix is physically indefinite; it only means that this particular inexpensive certificate cannot be used.

Table 7 Interpretation of local buckling assessment states

Outcome	What the calculation establishes	How it is used
Converged local RF	The implemented nested-space change criterion passes	Eligible finite RF, with method and refinement history
One hundred term approximation	A finite reduced Ritz subspace was re-solved	Eligible approximate RF, explicitly not converged; capacity may be high
Certified non governing case	A conservative lower bound exceeds a different converged governing RF	The unresolved case can be excluded from governing without inventing its eigenvalue
No demand marker 99	The specific complete assessed load field has no destabilizing demand	Sentinel, excluded from comparison with finite calculated RFs
Unassessed	Required geometry, recovery, material law or positive root is unresolved	No invented passing RF; known minima remain diagnostic

All finite RFs used in selecting a governing case retain their uncapped numerical values. Display saturation at 999 is a presentation choice. The no-demand value 99 is a separate sentinel and never takes precedence over a genuine finite result simply because it is numerically smaller. Those conventions must be retained when interpreting exported tables or building a separate sizing workflow from them.

13 Case eligibility coverage and the combined property result

13.1 Only a supported physical case can supply demand

The Stress Report explicitly identifies the analysis type. Current physical SOL101 and converged full-load SOL106 variants are eligible. Unknown stored/archive provenance, a missing physical static state, unavailable results, historical formulations, incompatible model signatures or meshes, recorded validation issues and failed or partial analysis states are rejected. A nonlinear case additionally needs a full-load convergence record. A partial nonlinear comparison is not treated as a completed limit case simply because it contains some stresses.

Table 8 Eligibility is checked before a property or panel demand is assessed

Source or state	Stress Report treatment	Reason
Current matching SOL101	Eligible when recovery is complete	Physical static demand
SOL106 converged at full load	Eligible with proportional demand estimate label	Physical state exists but factored nonlinear equilibrium is not recomputed
SOL103 normalized modes	Excluded	Amplitude is arbitrary
SOL105 parent and its preload variants	Outside this report envelope	Separate global buckling result scope
Stored archive lacking provenance	Excluded	Analysis and recovery contract cannot be verified

Source or state	Stress Report treatment	Reason
Failed partial stale or mismatched case	Excluded and relevant coverage withheld	Cannot represent the selected complete physical case

The SOL105 scope deserves care. Its static preload can be a physical state that the direct material viewer can inspect, but the SOL101 and SOL106 Stress Report does not add a SOL105 parent or its preload variants to its multi-case strength envelope. This is a deliberate reporting boundary. It does not change the material formula or turn a normalized buckling mode into a stress case.

13.2 Missing data is not zero loading

Each selected case has an expected set of physical members. Every required EID must provide the needed stress, strain face or resultant. A selected case that never returned and a relevant case excluded because it failed or no longer matches the model keep the coverage incomplete. Missing element recovery is not assigned zero merely to make a mean or RF finite.

Different criteria need different data. Euler buckling can use signed section axial force while a VM material envelope also needs bending, shear and torsion. A plate needs a complete signed membrane tensor in a common frame. A laminate strain check needs both physical faces of every eligible ply. A member can therefore have a complete result for one criterion and a valid unavailable reason for another without a contradiction.

13.3 Membership and minimum RF

The combined Minimum RF calculation associates every physical criterion with its actual member element IDs. A stringer column containing tributary skin can involve several property regions, and a panel can share a property with another panel. The same governing physical RF is assigned to every property it actually involves, with its shared scope identified. A property ID is a material or section assignment; it is not automatically a complete restrained panel or column.

$$RF_{p,\min} = \min_{m,c,q \in \mathcal{A}_p} RF_{m,c,q} \tag{90}$$

Minimum over all applicable complete failure modes, cases and physical scopes involving a property; only assigned when required coverage is complete.

The index set in this equation contains the applicable physical checks, load cases and scopes associated with the property. A complete combined RF requires every applicable required assessment to be complete. If another required criterion is missing or its membership cannot be verified, the property is marked unassessed even when one complete check already has RF below one. The code retains that smaller known value as partial information but does not promote it to a complete minimum envelope.

Generated models are also checked for missing criterion metadata and uncovered physical component elements. Older meshes can retain valid stresses while lacking the later panel or column membership definitions. The software then requests mesh regeneration rather than interpreting absence of metadata as proof that a failure mode is inapplicable. For imported models, generated rib-bay or plate boundaries are not inferred from a PID, so their available envelope can be material-only.

13.4 No demand and display limits

A complete stability check with no destabilizing demand can return the display sentinel RF 99. Complete zero-demand principal-strain checks use the same flag, and the combined layer can classify complete zero-demand material checks in this way. In the standalone VM property report, zero stress instead has no finite material RF. These presentations must be read with their noDemand metadata rather than as calculated finite capacities.

A sentinel does not cap the physical margin of another active check. If one complete mode has no demand and another has a genuine finite RF of 120, the latter governs; the program does not choose the numeric sentinel 99 as if it were a smaller physical capacity. Only when all applicable complete checks have no demand does the combined result use 99.

Similarly, the RF display cap of 999 does not replace the uncapped values used to select the governing case or criterion. A displayed 999 means that the finite calculated ratio exceeds the presentation cap, whereas 99 accompanied by no-demand metadata represents a different condition. A property at minimum gauge is a sizing constraint, not an additional failure mode or a modification of the underlying RF.

13.5 Retained and estimated assessments

Assessment policy and allowables are part of the postprocessing identity, while the physical FE model and mesh have their own signatures. Policy-only changes can reuse the physical solved state. When a compatible property edit is used for manual sizing, the program can update section capacity and demand using retained internal loads. Such an estimate holds the internal load distribution fixed. It cannot predict redistribution, changed global displacement or a new nonlinear equilibrium; the fresh analysis remains the verification of the edited structure.

The report and exports retain completeness, source, policy and approximation information so that a number can be traced back to the state and physical idealization that produced it. A critical case or failure-mode label without those accompanying conditions is insufficient to determine whether a property has actually been fully assessed.

14 The native whole model SOL105 buckling calculation

14.1 Preload and structural geometric stiffness

The native SOL105 calculation is separate from the local analytical plate and member assessments described earlier in the report. It first obtains the selected STATSUB static preload state. The full-model initial-stress geometric stiffness is assembled from that solved state through the relevant element formulation. The geometric stiffness is not reconstructed by merely reading a colored VM contour or applying the local Euler formula to every bar.

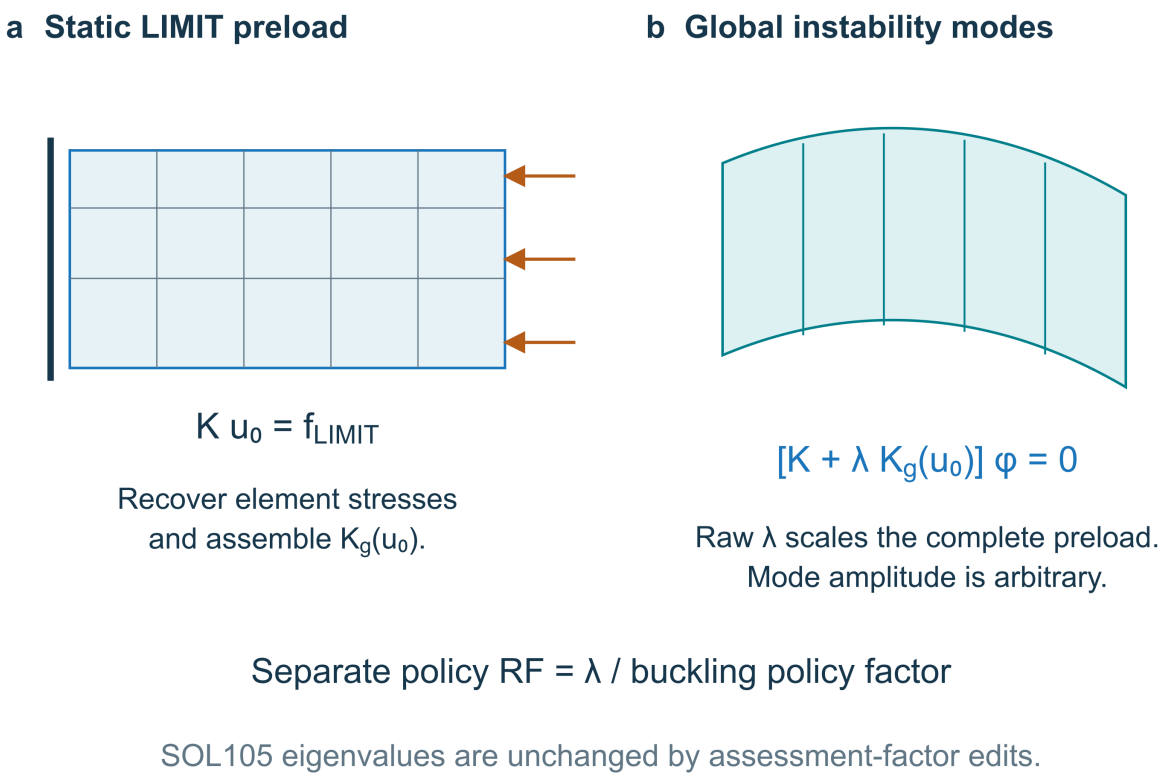


Figure 14: Whole model buckling and its preload

The native global pencil uses the assembled structural stiffness and geometric stiffness of a solved physical preload. Its normalized eigenmode is a perturbation shape, not a physical postbuckling deformation.

$$\left(\mathbf{K} + \lambda \mathbf{K}_g\right)\boldsymbol{\phi} = \mathbf{0}, \qquad \mathbf{B} = -\mathbf{K}_g, \quad \mathbf{K}\boldsymbol{\phi} = \lambda \mathbf{B}\boldsymbol{\phi} \tag{91}$$

Native global SOL105 pencil, distinct from local analytical plate/member systems.

The selected buckling SPC and MPC definitions are resolved independently of the preload case. The physical preload is retained in its GRID coordinate layout, while the perturbation problem is pulled back through its own constraint mapping. This matters when an authored buckling case intentionally has different perturbation boundary conditions. It is not correct to assume that every static support selection is automatically copied into the eigenproblem.

Lambda multiplies the complete analyzed LIMIT pattern: aerodynamic or prescribed actions and all enabled inertia, thrust and other loads present in that preload. It is not a multiplier of lift alone and is not itself aircraft g. A mode vector has arbitrary normalization, so multiplying its displayed amplitude does not predict the physical displacement at the critical load.

14.2 Extraction methods and request handling

The solver constructs the constrained free-coordinate pencil and retains the requested range and mode count. Eligible small full-spectrum requests can use dense inverse or generalized methods. Larger problems use factorized inverse or shift-invert Krylov operators. The backend is input-dependent and recorded in the diagnostics. It would therefore be inaccurate to state that every JFEM buckling problem uses a single dense solve, Arnoldi calculation or Lanczos calculation.

$$\mathbf{K}^{-1}\mathbf{B}\boldsymbol{\phi} = \theta\boldsymbol{\phi}, \qquad \lambda = \frac{1}{\theta} \tag{92}$$

The unshifted inverse operator maps a nonzero reciprocal eigenvalue theta to the original buckling multiplier.

$$(\mathbf{K} - \sigma\mathbf{B})^{-1}\mathbf{B}\boldsymbol{\phi} = \theta\boldsymbol{\phi}, \qquad \lambda = \sigma + \frac{1}{\theta} \tag{93}$$

The shifted inverse operator changes spectral targeting, not the original physical pencil. Every candidate is checked after back transformation.

At zero shift the largest-magnitude transformed eigenvalues correspond to small absolute critical multipliers. With a nonzero shift, the transformed operator concentrates extraction near the selected part of the spectrum. The elastic or shifted system is factorized; no explicit large inverse matrix is required. Cholesky can be used when the elastic constrained stiffness is positive definite, with other supported factorization paths used where appropriate.

The ordinary transformed operator K inverse B is not automatically symmetric in the Euclidean inner product, so the Krylov path can use Arnoldi. An optional Cholesky congruence provides a genuinely symmetric equivalent operator for Lanczos. Repeated eigenvalues require enough independent subspace directions; selected symmetric and block recovery routes and request-completeness checks address this issue. These are extraction safeguards, not changes to the physical elastic or geometric operators [8].

14.3 Verify an eigenpair in the original pencil

Convergence of the transformed iterative operator alone is not sufficient. After mapping a candidate back to its physical eigenvector, the implementation calculates the residual of the original generalized pencil. This catches loss of accuracy that can be hidden by inverse transformations, a large critical factor or cancellation near a shift. The denominator below is the maximum of the two physical terms and a machine-scale floor, rather than the sum used by some alternative residual conventions.

$$\rho = \frac{\|\mathbf{K}\boldsymbol{\phi} - \lambda\mathbf{B}\boldsymbol{\phi}\|}{\max\left(\|\mathbf{K}\boldsymbol{\phi}\|, |\lambda|\|\mathbf{B}\boldsymbol{\phi}\|, \epsilon_{machine}\right)} \tag{94}$$

Exact original generalized-pencil residual used by the implementation; denominator is a maximum, not the sum often used in other normalizations.

The default original-pencil acceptance tolerance is 1e-6, with the resolved numerical request and diagnostics determining the actual route. Singular shifts can be moved outward by controlled jitters, and range requests carry additional checks on recovered roots and completeness. An eigenvalue list should therefore be interpreted together with its accepted residuals and request-completion information, especially for closely clustered or repeated modes.

14.4 The policy RF is a separate presentation

$$RF_{SOL105,policy} = \lambda / f_b \tag{95}$$

Derived policy RF; raw eigenvalues and sensitivities stay unchanged.

Changing the buckling policy factor changes this derived assessment without changing raw native lambda, eigenvectors, ordering or sensitivity objectives. With a raw lambda of 1.5 and default policy one, the RF is 1.5; with policy 1.2 it is 1.25. The strength safety factor does not enter the calculation. Local plate and isolated-member RFs may differ from a global eigenvalue because they solve different idealized structures with different boundary, load-field and kinematic assumptions.

15 Worked buckling examples and interpretation

The examples in this chapter are manufactured method checks, not aircraft validation results. Their inputs are deliberately simple enough to reproduce independently. The current JavaScript modules were called directly, and the uniform isotropic plate result and combined-section Euler stiffness were checked against independent closed-form or enumerated calculations. No new FE equilibrium solve or comparison with an external solver was performed.

15.1 Isotropic skin panel under uniform compression

Take Young modulus 70 GPa, Poisson ratio 0.33, thickness 1.5 mm, equivalent length 0.4 m and width 0.12 m. The LIMIT membrane field is 10 kN/m compression along the panel length, with zero transverse force and shear. This is an uncoupled uniform case, so the code uses its exact integer-mode route. An independent enumeration of the first one hundred values in each modal direction finds the same minimum.

$$D_0 = \frac{70 \times 10^9 (0.0015)^3}{12(1 - 0.33^2)} = 22.09348 \tag{96}$$

Isotropic bending rigidity D0 = 22.09348 N·m for the manufactured panel, using SI inputs.

Table 9 Uniform compression method check

Quantity	Result
Governing sine mode	m = 3 and n = 1
Critical axial membrane resultant	61.24540511 kN/m
RF with buckling policy 1	6.12454051
RF with buckling policy 1.2	5.10378376

The three longitudinal half waves are a useful reminder that the first shape is not always the critical shape. Replacing the integer search with a fixed one-by-one mode would give a different resistance for this aspect ratio. Changing the material strength safety factor has no effect on any number in this table; changing the buckling policy divides the RF by that policy while leaving the physical critical resultant unchanged.

15.2 Biaxial compression and shear on the same panel

Retain the preceding geometry and bending law, but add transverse compression of 2.5 kN/m and signed shear of 3.5 kN/m to the axial 10 kN/m compression. The coupled Ritz calculation gives RF 4.45035932 with its convergence criterion satisfied. The lower RF follows from the full coupled geometric matrix, not an assumed empirical interaction equation. It is the factor multiplying the complete prescribed three-component load pattern.

$$\mathbf{C}_{\mathrm{LIMIT}} = (10,\ 2.5,\ 3.5)\ \mathrm{kN/m}, \qquad \mathrm{RF}_{\beta=1} = 4.45035932 \tag{97}$$

Compression-positive tensor components of the manufactured combined-load case.

With policy 1.2 the same computed critical pattern corresponds to RF 3.70863277. The two calculations describe the same elastic idealized resistance, but compare it with different allowed assessment load levels. Neither value contains the strength factor 1.5. The field direction, including the sign of shear, remains unchanged by this policy rescaling.

15.3 A web with zero mean force and local compression

Divide the equivalent rectangle into two equal-height cell bands. Give one band axial compression of 10 kN/m and the other equal axial tension, with zero transverse load and shear. The signed area mean is zero. A uniform mean-only plate assessment would consequently produce the no-demand sentinel 99 for that averaged field.

The spatial web method instead integrates both cell tensors and returns a finite converged RF of 15.73996870. This result demonstrates why the web method cannot be described merely as a check of its displayed average stress. The compressed subregion can destabilize a mode while the tensile subregion stabilizes another part of the same displacement field. Their contribution depends on the mode gradients within each cell, not only on the area average.

Table 10 The same zero mean does not imply the same instability model

Representation	Result and meaning
Uniform averaged tensor equal to zero	RF 99 marker for no destabilizing averaged demand
Two signed spatial web cells	Converged spatial RF 15.73996870 at policy 1

15.4 Combined stringer and skin Euler section

Use a beam T section with flange width 30 mm, total height 32 mm, flange thickness 2 mm and web thickness 2 mm. Add a 120 mm wide skin strip of thickness 1.5 mm, using the same 70 GPa modulus. Set the skin reference plane at zero, the flange centroid at minus 1 mm and the free-web centroid at minus 17 mm. These are the exact nonoverlapping flange and web rectangles used for the numerical section example; their eccentricity relative to the skin is retained.

The elastic centroid is minus 3.6 mm relative to the skin plane. The independent parallel-axis inertia matches the section module. With full unbraced length 0.4 m and Euler coefficient one, the critical load is 78.00704000 kN. A LIMIT compression of 5 kN gives RF 15.60140800 at policy 1, or RF 13.00117333 at policy 1.2. Splitting the 0.4 m member into more mesh bars would add section stations but would not change that complete unbraced length.

$$\mathrm{RF} = \frac{78007.0399977}{\beta\, 5000} = \begin{cases} 15.60140800, \beta = 1 \\ 13.00117333, \beta = 1.2 \end{cases} \tag{98}$$

A cutwise Euler resistance compared with the independent buckling-policy demand.

15.5 Local free web and policy examples

For a clear blade height of 30 mm and web thickness 1.5 mm, with modulus 70 GPa and Poisson ratio 0.33, the long-outstand law gives critical stress 68.64660039 MPa. A signed mean LIMIT compression of 50 MPa gives blade RF 1.37293201 at policy 1, or RF 1.14411001 at policy 1.2. This is a local plate comparison; the full rib-bay length and column EI do not enter it.

Finally, a native global buckling eigenvalue of one gives policy RF one at the default policy. A native eigenvalue of 1.5 gives RF 1.5 at policy 1 and RF 1.25 at policy 1.2. These are direct consequences of the stated demand convention. The native eigenvalue is not rewritten when the user changes a sizing factor.

Table 11 The requested LIMIT to policy RF convention

LIMIT critical multiplier	Buckling policy	Displayed policy RF
1	1	1
1.5	1	1.5
1.5	1.2	1.25

The full numeric inputs, unrounded intermediate results, convergence histories and reproducible module calls are retained in the private report build material. These examples exercise the implemented formulas and their units. They do not demonstrate that the simplified support, homogenization or averaging assumptions are appropriate for every aircraft component. That engineering judgment remains separate from checking that an equation has been evaluated correctly.

16 Fixed load sizing estimates and engineering interpretation

16.1 What remains fixed during a property edit

A sizing edit can reuse the internal forces recovered from the last compatible analysis while rebuilding the affected material and section data. This provides an immediate estimate of the local effect of a new thickness or section dimension. The baseline FE result objects and physical load pattern remain unchanged. The projection is accepted only when the retained topology, region membership, section-update certificate and required recovery are compatible.

For a simple axial member, the new stress can be recomputed from the retained axial force and the edited area. For a shell, a retained membrane resultant divided by the edited thickness provides the corresponding membrane stress. These identities explain why the estimate is based on force resultants rather than simply reusing the previous stress value.

$$\sigma'_{a,L} = \frac{N_L}{A'}, \qquad \sigma'_{x,L} = \frac{N_{x,L}}{t'} \tag{99}$$

A prime denotes edited section data. The retained quantities NL and Nx,L are force and force per unit width, respectively. Actual implementation eligibility and stress recovery remain criterion-specific.

Bending, shear and laminate updates require more information than these elementary axial identities. The current section or laminate law is reconstructed and the supported resultants are used to recover a new demand. Coverage is checked again. An incompatible geometry edit, unsupported material law or missing resultant cannot be repaired by a generic thickness-ratio rule.

16.2 Capacity scaling differs from demand scaling

$$D = \frac{Et^3}{12(1-\nu^2)}, \qquad N_{x,\mathrm{cr}} \propto D \tag{100}$$

For a fixed isotropic plate geometry, bending rigidity scales with t^3. At fixed membrane force per width, the corresponding critical-load RF scales with t^3, not with t^2.

The often-quoted t^2 critical-stress relation and the t^3 critical-resultant relation are consistent because stress is membrane resultant divided by thickness. An RF update must keep its force or stress reference clear. Changing a stringer web height, attached flange, material, offset or tributary skin thickness changes several section terms together. The weakest bending axis can rotate; there is no universal one-parameter power law for the combined column.

$$P'_{\mathrm{cr}} = \frac{c\pi^2 (EI)'_{\min}}{L^2}, \qquad RF'_b = \frac{P'_{\mathrm{cr}}}{f_b N_{L,c}} \tag{101}$$

This expression retains the LIMIT compression and member length while updating the section rigidity. Where the current code evaluates several cuts, the complete cutwise minimum remains necessary.

For a laminate, the updated stiffness is obtained from the reconstructed laminate law. A change of ply fractions, material or offset cannot generally be represented by a single isotropic thickness multiplier. The free-membrane-relaxation bending approximation, plate basis and homogenization assumptions described earlier still apply after the edit.

16.3 Why a new analysis is still needed

A stiffer or softer property changes the equilibrium solution and can redistribute loads to neighboring components. It also changes structural mass and, when enabled, inertia loading. A complete rerun updates those effects. The rapid estimate intentionally freezes the retained internal loads; it is therefore a local sizing indication rather than verification of the edited aircraft under a newly equilibrated load state.

The RF table, selected failure-mode field, critical-case field and element inspection use the same current assessment context. The RF below one filter follows the recalculated numerical RF, so a region that rises above one disappears from that filtered presentation. This visibility change does not constitute a new FE solve. Labels and source status identify a retained or estimated result.

Minimum-gauge coloring is a sizing aid. A yellow cell can identify a passing region whose relevant dimensions are already at their recorded minima. It does not change the calculated RF or add a new capacity criterion. A truncated plate result, incomplete assessment or unsupported check must still be interpreted from its own status.

16.4 Practical reading sequence

First confirm that the intended load cases are eligible and that the physical source is current or explicitly retained. Then confirm the LIMIT convention and the two assessment factors. Read the governing failure mode and physical scope, not only the property label. Inspect recovery coverage and any plate approximation or lower-bound certificate. After a sizing edit, use the estimate to guide further changes, then rerun the analysis to obtain redistributed internal loads and reassess the complete applicable set.

17 Implementation traceability and verification

The reviewed implementation is a working-tree snapshot, including the separate LIMIT, ULTIMATE and buckling-policy changes. A Git commit identifier alone is not sufficient to reproduce these methods when uncommitted source files differ from that commit. The source catalogue in the appendix records function locations, roles and file fingerprints for the reviewed bytes.

17.1 Calculation checks and physical validation answer different questions

The worked examples in this report exercise the actual JavaScript assessment modules and independent closed-form or discrete-mode oracles. They check units, algebra, load-policy scaling, weak-axis section stiffness and the plate eigensystem. They are manufactured method checks, not new FE analyses or comparisons with an external experimental data set.

The existing regression evidence includes backend policy and import contracts, aircraft lift-matching tests, TOML round trips, calculator tests and actual browser workflows. It verifies that an assessment-factor edit leaves physical FE loads, cards, result arrays and signatures unchanged while updating RFs. The browser tests deliberately use specified synthetic recovery on a genuinely generated mesh to test mapping, coverage and presentation.

Table 12 Representative implementation verification

Area	Question checked	Evidence type
Policy and loading	Are safety and buckling factors independent, positive and separate from the physical case multiplier?	Julia contracts and module tests
Strength and statistics	Do stress/strain selection, equal-element averaging, percentile ties and case envelopes follow the recorded rules?	Calculator tests and numerical examples
Local buckling	Do section stiffness, signed membrane terms, convergence and policy scaling follow the implemented equations?	Module checks and independent plate/EI oracles
Complete application	Do nonempty checks for all eight buckling families update consistently in tables, fields and inspection?	Generated RF-rich browser fixture
Asynchronous assessment	Can canceled or superseded workers publish an old factor or stall the next assessment?	Deterministic protocol checks and delayed-worker browser tests
Persistence and exports	Do saved Studies, imported controls and exported tables preserve the actual policy and status?	Study/TOML, browser and export checks

17.2 Interpretation of the evidence

Agreement with a closed-form isotropic plate or Euler example verifies a controlled mathematical case. It does not establish that the equivalent rectangle, tributary section or chosen fixity coefficient represents every real wing detail. Conversely, a well-converged numerical eigenvalue can still have modelling error if a physical restraint, material law, mesh gradient or load path is represented inadequately. The report therefore states both numerical convergence and physical idealization assumptions.

The bibliography identifies primary sources for classical laminate mechanics, Euler columns, plate energy methods, shear flow and numerical eigensolvers. These sources explain the underlying theory. The application-specific averaging, membership, coefficient choices, finite basis, scope exclusions and display conventions are established by the code catalogue and its accompanying source audit.

A Symbols units and numerical status

Symbols follow the local context defined with each equation. In particular, A denotes section area in a beam expression and the extensional stiffness matrix in a laminate expression. Local plate G is compression-positive; the native whole-model geometric stiffness uses its own tensile-positive assembly convention.

Table 13 Symbols and units

Symbol	Meaning	Units or interpretation
RF	Reserve factor	dimensionless
s	Strength safety factor	dimensionless
f_b	Buckling policy factor	dimensionless
m_c	Physical case multiplier	dimensionless
n	Aircraft maneuver load factor	dimensionless
M	Complete-aircraft mass	kg
g	Standard gravity	m/s^2
u	FE displacement and rotation vector	m and rad
σ and τ	Normal and shear stress	Pa; MPa in generated-model displays
ε and γ	Normal and engineering shear strain	dimensionless
κ	Shell curvature	1/m
N_x N_y N_{xy}	In-plane force resultants per unit width	N/m
M_x M_y M_{xy}	Shell bending/twisting moment resultants per unit width	N
N	Member axial force where used as a scalar	N
E	Young modulus	Pa
A	Cross-sectional area in a member formula	m^2
A B D	Laminate extensional coupling and bending matrices	N/m; N; N·m
D*	Free-membrane-relaxed effective bending matrix	N·m
EI	Elastic flexural rigidity	$N·m^2$
L	Recorded actual unbraced member path length	m
a and b	Equivalent plate length and width	m
t and h	Thickness and clear blade height	m
c	Engineering capacity or restraint coefficient	dimensionless
K_β	Local plate bending stiffness in a Ritz basis	N/m for displacement amplitudes in m
G_p	Compression-positive local plate geometric stiffness	N/m for displacement amplitudes in m
K and K^g	Native FE elastic and geometric stiffness matrices	DOF-dependent consistent units
λ	Native global buckling multiplier or local critical load-pattern amplitude	dimensionless

Symbol	Meaning	Units or interpretation
μ	Reciprocal local critical load amplitude	dimensionless
99	Displayed complete no-destabilizing-demand marker	sentinel
999	Maximum displayed calculated RF	display cap
s_N	Positive demand magnitude used to normalize the signed plate membrane field	N/m
Stored K_b	Mode-area-normalized code bending matrix	N/m³
Stored normalized G	Mode-area and demand-normalized code matrix	1/m²
μ_hat	Largest eigenvalue of the code-normalized plate problem	m/N; physical μ = s_N × μ_hat

B Implementation source catalogue

The source paths below are relative to the integrated repository root. The line anchors identify the reviewed snapshot and may move in later revisions. Function names and file fingerprints provide the corresponding implementation trace.

component_buckling.jl

APPS/JFEM-Boxer/src/component_buckling.jl

Line 348 component_buckling_payload — Scope of the implemented buckling assessments

Line 255 component_column_buckling_payload — Bare spar cap and rib stiffener Euler columns

Line 297 component_column_buckling_payload — Bare spar cap and rib stiffener Euler columns

Line 309 component_column_buckling_payload — Bare spar cap and rib stiffener Euler columns

Line 244 component_rf_blade_bar — Local free blade instability of stringers and rib stiffeners

Line 84 component_rf_compartments — Skin and web regions are physical compartments rather than property IDs

Line 105 component_rf_web_geometry! — Skin and web regions are physical compartments rather than property IDs

Line 203 component_web_buckling_payload — Skin and web regions are physical compartments rather than property IDs

jfem_run.jl

APPS/JFEM-Boxer/src/jfem_run.jl

Line 1356 append_solid_results_payload! — Imported native volume EIDs, centroid contour values and full available stress row; no nodal averaging.

Line 1235 SHELL_FORCE_FIELDS and BAR_FORCE_FIELDS — Canonical payload force fields and units.

Line 1276 complete_stringer_stresses! — Complete geometric polygon-vertex bending stresses for T/BAR generated members and valid imported sections.

Line 1398 element_results_payload — EID-specific recovered records, frame rotation, explicit units, no sharing or smoothing between adjacent elements.

shell_frames.jl

APPS/JFEM-Boxer/src/shell_frames.jl

Line 10 shell_geometric_frame — Native geometric axes from element connectivity; quadrilateral default diagonal frame.

Line 49 shell_material_frame — Generated skin/stringer, rib and spar material/result reference axes.

Line 134 shell_result_rotation — Generated element geometric-to-material rotation; imported models retain native axes.

Line 143 rotate_shell_results! — Tensor rotation for N/M/stress and gamma-correct strain rotation; vector rotation for Q.

skin_buckling.jl

APPS/JFEM-Boxer/src/skin_buckling.jl

Line 73 skin_buckling_payload — Scope of the implemented buckling assessments

Line 188 skin_buckling_payload — Skin and web regions are physical compartments rather than property IDs

Line 213 skin_buckling_payload — Skin and web regions are physical compartments rather than property IDs

Line 17 skin_laminate_abd — Laminate bending rigidity for the plate assessments

Line 39 skin_effective_bending — Laminate bending rigidity for the plate assessments

Line 49 skin_stress_transform — Signed membrane recovery and the difference between skin and web loading

stringer_blade.jl

APPS/JFEM-Boxer/src/stringer_blade.jl

Line 12 stringer_blade_payload — Local free blade instability of stringers and rib stiffeners

Line 51 stringer_blade_payload — Local free blade instability of stringers and rib stiffeners

stringer_euler.jl

APPS/JFEM-Boxer/src/stringer_euler.jl

Line 271 stringer_euler_payload — Scope of the implemented buckling assessments; Physical stringer column and tributary skin membership

Line 327 stringer_euler_payload — Physical stringer column and tributary skin membership

Line 337 stringer_euler_payload — Physical stringer column and tributary skin membership

Line 357 stringer_euler_payload — Physical stringer column and tributary skin membership

Line 365 stringer_euler_payload — Physical stringer column and tributary skin membership

Line 383 stringer_euler_payload — Physical stringer column and tributary skin membership

Line 51 euler_rectangle — Column section stiffness and the weakest bending axis

Line 194 euler_beam_parts — Column section stiffness and the weakest bending axis

Line 61 euler_shell_section — Laminate strips and oblique section cuts

Line 92 euler_material_q — Laminate strips and oblique section cuts

Line 116 euler_laminate_axial — Laminate strips and oblique section cuts

Line 147 euler_strip_matrix — Laminate strips and oblique section cuts

Line 218 euler_shell_part — Laminate strips and oblique section cuts; Column compression is recovered at cuts rather than summed along a path

stringer_euler_spatial.jl

APPS/JFEM-Boxer/src/stringer_euler_spatial.jl

Line 85 euler_section_candidates — Physical stringer column and tributary skin membership

assessment_load_policy.js

APPS/JFEM-Boxer/web/assessment_load_policy.js

Line 6 DEFAULTS/read — Default strength factor 1.5 and buckling factor 1, finite positive validation.

Line 29 sample — Apply strength safety factor once to utilization while retaining LIMIT and ULTIMATE demand fields separately.

Line 8 read — Independent load policy and restraint coefficients

buckling_outputs.js

APPS/JFEM-Boxer/web/buckling_outputs.js

Line 24 render — No demand markers and display saturation

Line 4 policyReserveFactor — Whole model SOL105 and its policy RF are separate from local checks

component_buckling_data.js

APPS/JFEM-Boxer/web/component_buckling_data.js

Line 130 build — Scope of the implemented buckling assessments; Bare spar cap and rib stiffener Euler columns

Line 107 assessWeb — Independent load policy and restraint coefficients

Line 13 normalizedBounds — Skin and web regions are physical compartments rather than property IDs

Line 21 validateTiling — Skin and web regions are physical compartments rather than property IDs

Line 102 spatialDemand — Signed membrane recovery and the difference between skin and web loading

Line 28 integralCos / integralSin / integrals — Rayleigh Ritz plate formulation; Exact cell integration for spatial web fields

Line 35 spatialMatrices — Rayleigh Ritz plate formulation; Exact cell integration for spatial web fields

Line 49 criticalForSpatialBasis — Exact cell integration for spatial web fields; The numerical eigenvalue algorithm and signed tensile stabilization

Line 73 solveSpatialPlate — Convergence and the uncoupled exact integer mode route

Line 87 solveSpatialPlate — Explicit one hundred term approximations after nonconvergence

Line 61 spatialLowerBound — Conservative lower bounds certify only non governing unresolved cases

Line 117 assessWeb — Conservative lower bounds certify only non governing unresolved cases

Line 55 nonDestabilizing — No demand markers and display saturation

Line 74 solveSpatialPlate — Maximum spatial demand magnitude; normalized cells line76; final-only scaling line94.

Line 115 assessWeb — History fixity and policy scaling without additional division by magnitude.

euler_column_data.js

APPS/JFEM-Boxer/web/euler_column_data.js

Line 55 criticalLoad — Independent load policy and restraint coefficients

Line 10 matrixPart — Column section stiffness and the weakest bending axis

Line 28 section — Column section stiffness and the weakest bending axis

Line 63 forceTerm — Column compression is recovered at cuts rather than summed along a path

Line 76 axialForce — Column compression is recovered at cuts rather than summed along a path

Line 94 assessColumn — Column compression is recovered at cuts rather than summed along a path

Line 115 assessColumn — No demand markers and display saturation

material_allowables.js

APPS/JFEM-Boxer/web/material_allowables.js

Line 6 DEFAULT_STRESS — Al 2024 generated-SI design default 300 MPa, distinct from certified material allowables.

Line 35 strainSample — Tensile/compressive principal strain from each explicit ply-face engineering strain tensor.

Line 47 equivalentStressSample — Both PSHELL stress faces required; recursively select explicitly recovered VM peak; no axial-stress substitute for bars.

Line 62 assessElement — Material-specific allowable checks, homogeneous-MAT8 laminate eligibility and complete ply-face coverage.

minimum_rf_data.js

APPS/JFEM-Boxer/web/minimum_rf_data.js

Line 73 caseAssessments — Per-case criterion completeness, zero-demand sentinels and factor-aware material RF.

Line 91 build — Associate physical criteria to actual property members; all applicable complete criteria required for a complete combined RF.

Line 143 build governing minimum — Select minimum over complete applicable physical checks; retain known minimum without claiming complete assessment if another required check is missing.

Line 144 build — No demand markers and display saturation

skin_buckling_data.js

APPS/JFEM-Boxer/web/skin_buckling_data.js

Line 253 assessPanel — Independent load policy and restraint coefficients

Line 234 preparePanel — Laminate bending rigidity for the plate assessments

Line 241 panelDemand — Signed membrane recovery and the difference between skin and web loading

Line 32 buildMatrices — Rayleigh Ritz plate formulation

Line 86 basisPrepared — Rayleigh Ritz plate formulation

Line 16 cholesky — The numerical eigenvalue algorithm and signed tensile stabilization

Line 51 largestEigenvalue — The numerical eigenvalue algorithm and signed tensile stabilization

Line 80 transformGeometric — The numerical eigenvalue algorithm and signed tensile stabilization

Line 108 criticalForBasis — The numerical eigenvalue algorithm and signed tensile stabilization

Line 142 diagonalPlate — Convergence and the uncoupled exact integer mode route

Line 159 solvePlate — Convergence and the uncoupled exact integer mode route; No demand markers and display saturation

Line 121 truncateRitzMatrices — Explicit one hundred term approximations after nonconvergence

Line 206 solvePlate — Explicit one hundred term approximations after nonconvergence

Line 223 plateLowerBound — Conservative lower bounds certify only non governing unresolved cases

Line 268 assessPanel — Conservative lower bounds certify only non governing unresolved cases

Line 161 solvePlate — Demand magnitude and normalized signed direction; final/history scaling at lines167 and172.

stress_report_data.js

APPS/JFEM-Boxer/web/stress_report_data.js

Line 14 aggregate — Equal-element property statistics, linear percentile interpolation, and below-cutoff mean.

Line 31 caseReason — Explicit SOL101/SOL106 physical results only, matching current mesh/signature and complete convergence.

Line 48 caseList — Selected cases and physical variants, missing/relevant-excluded coverage.

Line 72 propertyAllowable — One MAT1 material and positive common stress allowable; PCOMP excluded from property VM report.

Line 94 principalStrainReport — Property-level worst homogeneous-MAT8 ply-face utilization across elements and cases.

Line 129 buildReport — Dispatch per failure mode and construct combined criterion reports.

Line 154 buildReport strength RF — RF=allowable/envelope/strengthSafetyFactor only with complete case and element coverage.

stringer_blade_data.js

APPS/JFEM-Boxer/web/stringer_blade_data.js

Line 7 COEFFICIENT — Local free blade instability of stringers and rib stiffeners

Line 10 criticalStress — Local free blade instability of stringers and rib stiffeners

Line 21 signedMean — Local free blade instability of stringers and rib stiffeners

Line 33 axialStress — Local free blade instability of stringers and rib stiffeners

Line 51 assessBlade — Local free blade instability of stringers and rib stiffeners

Line 82 assessBlade — No demand markers and display saturation

FEMKernels.jl

src/FEMKernels.jl

Line 6620 stress_strain_quad4 — Element recovery using actual shell constitutive and selected compatible, condensed, projected or ordinary formulation.

Line 6803 stress_strain_quad4 incompatible amplitudes — Internal Wilson-mode recovery alpha=-Kbb^-1 Kab^T ue and area-weighted GP fields in ordinary branch.

Line 9815 stress_strain_tria3 — Linear triangular membrane/curvature matrices and in-plane section recovery; MITC3 shear estimate.

JFEMSolver.jl

src/JFEMSolver.jl

Line 3331 SOL105 STATSUB orchestration — Retrieve physical preload state, select eigen MPC/SPC separately and assemble Kg.

Line 3450 solve_buckling call — Pass actual elastic/geometric pencil and explicit extraction request.

Line 3331 _solve_sol105 — Whole model SOL105 and its policy RF are separate from local checks

Line 3430 _solve_sol105 — Whole model SOL105 and its policy RF are separate from local checks

ModelBuilder.jl

src/ModelBuilder.jl

Line 22 laminate_plane_stress_qbar — Reduced orthotropic plane-stress constitutive matrix and transformed Qbar.

Line 757 PCOMP CLT integration — Integrate physical ply Qbar over z for A, B and D and retain original plies for face recovery.

assembly.jl

src/solver/assembly.jl

Line 5934 assemble_geometric_stiffness — Native full-model initial-stress geometric stiffness from solved preload.

beam_stress_envelope.jl

src/solver/beam_stress_envelope.jl

Line 19 beam_t_stress_envelope — Conservative equivalent-stress envelope for dimension-defined thin-walled MAT1 T sections, or dispatch to BAR rectangle.

Line 90 beam_rectangular_stress_envelope — Conservative solid rectangular section equivalent-stress envelope.

solve_case.jl

src/solver/solve_case.jl

Line 54 _assemble_applied_force — Resolve applied load set and thermal loads, rotate BASIC forces into GRID CD analysis axes, redistribute dependent RBE3 loads.

Line 182 _build_results_from_state — Build displacements, SPC reactions, element forces, stresses and strains from a physical static state.

Line 279 _build_results_from_state recovery dispatch — Shell recovery optionally uses corotational element distortions; bar/rod/spring/solid recovery dispatch follows.

Line 2514 _buckling_original_pencil_residual — Verify eigenpair in original generalized pencil, not only transformed Krylov operator.

Line 2782 solve_buckling — Native constrained generalized buckling eigensolver and request diagnostics.

Line 3102 attempt_shifted_buckling_search — Range-targeted shift invert using K-sigma B, B=-Kg, with singular-shift jitter.

Line 3715 dense buckling strategies — Dense inverse/generalized extraction for eligible bounded full-spectrum problems.

Line 3826 Krylov inverse iteration — K^-1 B transformed pencil; Cholesky or LU factorization, inverse transformed eigenvalues.

Line 4032 original pencil residual acceptance — Physical-pencil residual screening defaults to 1e-6; input-dependent backend and completeness diagnostics retained.

Line 3770 solve_buckling — Whole model SOL105 and its policy RF are separate from local checks

Line 3826 solve_buckling — Whole model SOL105 and its policy RF are separate from local checks

Line 4032 solve_buckling — Whole model SOL105 and its policy RF are separate from local checks

stress_recovery.jl

src/solver/stress_recovery.jl

Line 17 _recovery_principal_2d — Stable principal values of a symmetric 2D tensor; input third argument is tensor shear.

Line 42 _pshell_stress_fiber_distance — Default PSHELL fibers follow +/-T/2; explicit Z1/Z2 output distances remain explicit.

Line 53 _recovery_rotate_engineering_strain — Engineering-strain rotation with gamma12 rather than tensor epsilon12.

Line 69 _recovery_ply_face_rows — Bottom and top strain/stress in every original ply material frame, with completeness and frame guards.

Line 431 recover_shell_stresses! — Shell recovery orchestration, projected geometry, frame/offset/section-law branches.

Line 998 recover_shell_stresses! generalized output — Legacy strain z1 is membrane strain, z2 is negative curvature; explicit ply faces are separate.

Line 1111 recover_shell_stresses! output rows — Emit membrane/bending/transverse resultants and z1/z2 plane-stress entries.

Line 1298 recover_bar_stresses! — Frame construction, nodal coordinate and offset transformations, fixed-end-load subtraction, force/stress recovery.

Line 1320 frame_force_dict_from_local — Maps local beam nodal vector to signed section axial/shear/torque/end-moment resultants.

Line 1360 recover_bar_stresses! displacement frame — Transform GRID-CD displacements and rotations to beam axes, account for offsets and thermal strain.

Line 1434 recover_bar_stresses! axial and bending fields — Axial N/A is separate from end-point bending stresses; station-varying endpoints retain separate axial fields.

Line 1701 recover_solid_stresses! — Centroidal linear-elastic solid fields; ordinary CHEXA8 can additionally recover corners and enhanced-strain fields.

Reviewed file fingerprints

The full SHA256 values are recorded in source_snapshot.json beside the report source. The short values below identify the same reviewed files; the source manifest itself is also fingerprinted in the final build receipt.

Table 14 Reviewed source files

Repository relative file	SHA256 prefix
APPS/JFEM-Boxer/src/component_buckling.jl	5d3569da40fc2321
APPS/JFEM-Boxer/src/jfem_run.jl	5e72fe4d078af37f
APPS/JFEM-Boxer/src/shell_frames.jl	9ae843d79cdc00e3
APPS/JFEM-Boxer/src/skin_buckling.jl	3543ba66b1c46520
APPS/JFEM-Boxer/src/stringer_blade.jl	e57892f52ef225a4

Repository relative file	SHA256 prefix
APPS/JFEM-Boxer/src/stringer_euler.jl	73dfd43882680638
APPS/JFEM-Boxer/src/stringer_euler_spatial.jl	8e0667c82e84d5f9
APPS/JFEM-Boxer/web/assessment_load_policy.js	6b3d5863a82d6ff9
APPS/JFEM-Boxer/web/buckling_outputs.js	228200a7415c18f8
APPS/JFEM-Boxer/web/component_buckling_data.js	14185123c4a464ab
APPS/JFEM-Boxer/web/euler_column_data.js	4a3e0e7981b00936
APPS/JFEM-Boxer/web/material_allowables.js	b64c08d7243055d0
APPS/JFEM-Boxer/web/minimum_rf_data.js	9b281e3562e7be9e
APPS/JFEM-Boxer/web/skin_buckling_data.js	0fe81872970866e1
APPS/JFEM-Boxer/web/stress_report_data.js	52ec328c3adb8593
APPS/JFEM-Boxer/web/stringer_blade_data.js	1e88bb15ca0b9eec
src/FEMKernels.jl	c3c939bda8b442ec
src/JFEMSolver.jl	91d0b4f03e754855
src/ModelBuilder.jl	4395b102ab6b8046
src/solver/assembly.jl	41b921fa5c1bf19a
src/solver/beam_stress_envelope.jl	f58021f42f295cbf
src/solver/solve_case.jl	3ab5ad2bebafdf74
src/solver/stress_recovery.jl	7b7a2fab906c44e4

C Primary references

Classical references support the theoretical background. The implementation-specific recovery, averaging, eligibility, coefficients, finite bases and status rules are documented by the code catalogue. No external reference is used as evidence that every wing detail satisfies the assumptions of its idealized model. All links were checked on 3 November 2026.

[1] Alan T Nettles. Basic mechanics of laminated composite plates. NASA RP 1351. 1994.

https://ntrs.nasa.gov/citations/19950009349

[2] George Gerard and Herbert Becker. Handbook of Structural Stability Part I Buckling of Flat Plates. NACA TN 3781. 1957.

https://ntrs.nasa.gov/citations/19930084505

[3] Tomasz Wierzbicki. Plates and Shells Lecture Notes. MIT OpenCourseWare 2.081J and 16.230J. 2007.

https://ocw.mit.edu/courses/2-081j-plates-and-shells-spring-2007/1bd32d641a2b41ca518591d981405393_lecturenote.pdf

[4] David M Parks. Euler Bernoulli Beams Bending Buckling and Vibration. MIT OpenCourseWare 2.002 Lecture 2. 2004.

https://www.ocw.mit.edu/courses/2-002-mechanics-and-materials-ii-spring-2004/bc25a56b5a91ad29ca5c7419616686f7_lec2.pdf

[5] Mississippi State University Structural Analysis II. Shear Flow Analysis for Symmetric Beams.

https://www.ae.msstate.edu/tupas/SA2/chA14.7_text.html

[6] Mississippi State University Structural Analysis II. Shear Flow Analysis for Unsymmetric Beams.

https://www.ae.msstate.edu/tupas/SA2/chA14.8_text.html

[7] Mississippi State University Structural Analysis II. Torsion of Non Circular Bars.

https://www.ae.msstate.edu/tupas/SA2/chA6.4_text.html

[8] KrylovKit official documentation. Eigenvalue problems.

https://jutho.github.io/KrylovKit.jl/stable/man/eig/

[9] Julia official documentation. Linear Algebra standard library.

https://docs.julialang.org/en/v1/stdlib/LinearAlgebra/